

Semantic Matching in Medieval Latin: A Benchmark, Layerwise Analysis, and Geometric Repair

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Abstract

Medieval canon law before the twelfth century survives in many compilations, whose manuscripts differ in spelling and in the selection and arrangement of their elements. We introduce a Medieval Latin benchmark of 840 labelled canon-law fragments with 1,705 manuscript witnesses, and we define two tasks on it: pairwise duplicate detection and open-set directory routing. Witnesses to one fragment are hand copies of one exemplar and differ mostly in spelling, so a character n-gram baseline is as strong as any embedding configuration on this corpus; our question is where inside a pre-trained encoder the retrieval signal lives, why it disappears, and what recovers it. A layerwise analysis of six pre-trained models from three architectures separates two effects that are easy to conflate. Anisotropy is universal across the six models: every model has a layer where the mean cosine between randomly selected passages reaches 0.23 to 0.95, and no baseline layer in any model separates equivalent from non-equivalent passages by more than 0.29 in mean cosine. Catastrophic mid-depth collapse is not universal: the Latin and multilingual T5 encoders fall to AUROC 0.496, 0.538, and 0.654 in their middle layers, while LaBSE, Qwen3-0.6B, and KaLM-mini improve close to monotonically with depth. All-but-the-Top post-processing, fit only on training embeddings, repairs all six. Best-layer AUROC moves into a 0.971 to 0.987 band, the cosine gap rises to between 0.525 and 0.611, and the spread in directory accuracy at rank 1 across models falls from 39.3 points to 3.3. Under the repair, integrated gradients and retrieval-adapted MaRC give a narrow answer: ABTT improves leave-one-out rank faithfulness in every LaTa, PhilTa, and mT5-base model-method cell, while ERASER-style rationale metrics show mixed effects.

1 Introduction

Medieval canon law preserves early records of legislation and judicial opinions in Western Europe. Before the twelfth century it survives not as one stable text but in many compilations, whose manuscripts differ in spelling, wording, fragment boundaries, and transmission history. Deciding whether two passages express the same legal content despite those surface changes drives editorial work: identifying duplicates, analyzing witnesses, and tracing how a text was transmitted.

We study this problem as semantic matching on a benchmark of 840 labelled fragments of Medieval Latin canon law, each a canon or excerpt with one or more manuscript witnesses, 1,705 in all. Task A asks whether two passages express the same meaning. Task B turns the same embeddings into an open-set workflow: route a new passage to an existing directory, or mark it as new. Witnesses to one fragment are hand copies of one exemplar and differ mostly in spelling, so a character n-gram baseline matches the best embedding configuration on both tasks (Section 5); our question is not whether embeddings beat string matching but where inside a pre-trained encoder the retrieval signal lives, why it disappears, and what recovers it.

A final-layer vector cannot answer that, so we work layer by layer, and the layerwise view of six models separates two effects that are often reported together. Anisotropy is universal on this corpus: every model has a layer where pooled passages crowd into a narrow cone, with mean cosine between randomly selected passages reaching 0.23 to 0.95, and no baseline layer of any model separates equivalent from non-equivalent passages by more than 0.29 in mean cosine (Figure 1). A catastrophic mid-depth collapse of retrieval is not universal. LaTa and PhilTa fall to chance in their middle layers and mT5-base loses more than half of its above-chance separation, while LaBSE, Qwen3-0.6B, and

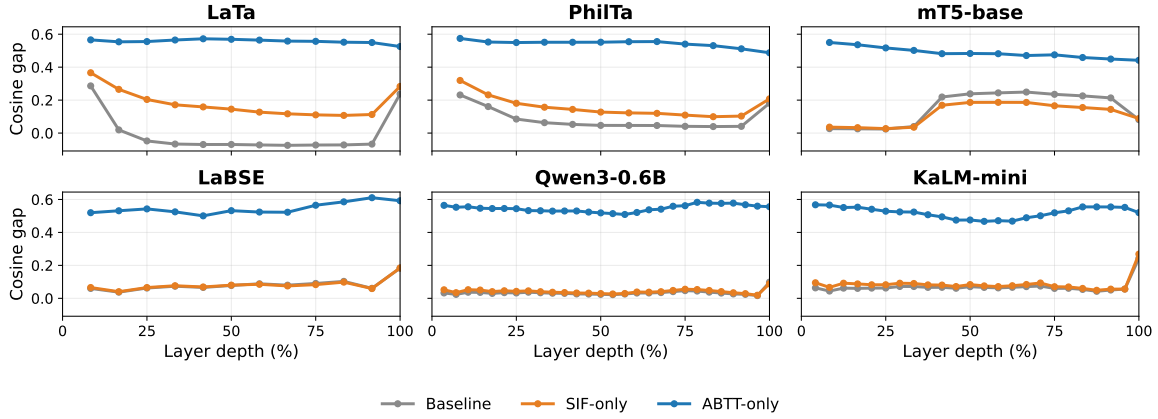


Figure 1: Cosine gap against layer depth for all six models: mean cosine of same-directory pairs minus mean cosine of different-directory pairs, the margin a threshold works with, so larger is better. Gray is baseline mean pooling, blue is ABTT-only with D tuned on train, orange is SIF-only. Baseline stays under 0.29 at every layer of every model, and under 0.11 for Qwen3-0.6B. Mid-depth in the top row, the LaTa and PhilTa baseline gap collapses toward zero with their retrieval; the mT5-base gap rises as its ranking fails, which only AUROC (Figure 2) shows. ABTT lifts all six near 0.5 at every depth. SIF-only helps LaTa and PhilTa and leaves the other four near baseline.

KaLM-mini improve close to monotonically with depth.

Anisotropy motivates our use of All-but-the-Top (ABTT) post-processing (Mu and Viswanath, 2018). We apply it after pooling with no update to model parameters: fit the dominant principal components on training embeddings, remove them from train and test embeddings, then score by cosine. The repair, unlike the collapse, is universal: ABTT lifts every model into a narrow high-accuracy band on both tasks, and at each T5 encoder’s worst baseline layer it recovers pairwise AUROC to within 1.0 point of that model’s best ABTT layer. SIF weighting (Arora et al., 2017), which reweights tokens by frequency before pooling, corrects part of the same direction but helps only the collapsed models, and SIF-only never matches ABTT-only.

Geometry diagnoses collapse and recovery but does not select the retrieval layer: a train-only rule (Section 4.2) locks LaTa layer 7, PhilTa layer 1, and mT5-base layer 1 before any attribution metric is computed.

Finally, we compare local explanations of the score before and after the repair. Integrated gradients (Sundararajan et al., 2017) gives a gradient view of token importance and retrieval-adapted MaRC (Brinner and Zarrieß, 2023) a learned-mask view, with MaRC’s classification target replaced by the pairwise retrieval cosine; both explain a fixed query-candidate score and stay local diagnostics.

This paper makes four contributions: (1) we introduce a benchmark for semantic matching in Medieval Latin canon law, together with a routing

task that states the scholar’s decision directly: file an unidentified witness under a known fragment, or flag it as new; (2) we separate, across six models, universal anisotropy from the mid-depth retrieval collapse that only the three T5 encoders show, measuring both with top-PC dominance, effective rank, and cosine concentration; (3) we show that ABTT repairs the anisotropy on this corpus without updating parameters, compressing a 39.3-point spread in directory accuracy at rank 1 to 3.3 points, with the largest gains on the T5 encoders; and (4) we adapt token attribution to bi-encoder retrieval, comparing a MaRC-style learned mask with integrated gradients at predeclared retrieval-selected layers; on present evidence ABTT improves leave-one-out rank faithfulness while ERASER-style rationale metrics stay mixed.

2 Related Work

Anisotropy, layer geometry, and repair. Pre-trained spaces occupy narrow cones (Gao et al., 2019; Ethayarajh, 2019) where a few coordinates dominate cosine (Kovaleva et al., 2021; Timkey and van Schijndel, 2021), and corrections remove or rescale those directions (Mu and Viswanath, 2018; Su et al., 2021) or downweight frequent tokens (Arora et al., 2017). Information spreads across depth (Tenney et al., 2019; Liu et al., 2019), intermediate layers can beat the last (Skean et al., 2025), and multilingual work is rarer (Rajae and Pilehvar, 2022). Layerwise profiles put the mid-depth anisotropy peak in decoder-only models (Razhigaev et al., 2024); T5 enters that work with encoder

and decoder states concatenated (Godey et al., 2024), and its weak mean-pooled similarity is a final-layer result (Ni et al., 2022). Anisotropy is neither universal (Machina and Mercer, 2024) nor always harmful (Ait-Saada and Nadif, 2023), so we report geometry and retrieval separately, under a train-only layer rule.

Latin and classical-language NLP. Latin has dedicated encoders and encoder-decoders (Bamman and Burns, 2020; Riemenschneider and Frank, 2023a), and appears in multilingual and decoder-only models (Xue et al., 2021; Feng et al., 2022; Zhang et al., 2025; Hu et al., 2025). Shared tasks target tagging (Sprugnoli et al., 2020), retrieval work targets allusion (Riemenschneider and Frank, 2023b), and charter work semantic change (Liu et al., 2025); no general benchmark (Thakur et al., 2021; Muennighoff et al., 2023) covers medieval Latin retrieval.

Text reuse in digital humanities. Text-reuse systems match n -grams (Coffee et al., 2012), align reprinted newspapers (Smith et al., 2013), and add distributional semantics for allusive reuse (Manjavacas et al., 2019; Burns et al., 2021), ranking candidates in a closed collection. Our routing task is open-set: a witness may match no labelled fragment in the collection.

Faithfulness of token attributions. Integrated gradients (Sundararajan et al., 2017) and learned masks (Brinner and Zarri  , 2023) score tokens, and their faithfulness is debated (Jain and Wallace, 2019; Jacovi and Goldberg, 2020). Erasure protocols (DeYoung et al., 2020; Atanasova et al., 2020) target classification and retrieval explanation targets ranking (Anand et al., 2022); we apply both to a bi-encoder cosine score.

3 Task and Dataset

3.1 Task Definition

The benchmark defines two tasks over the same witnesses and the same pooled embeddings. They differ in what the system has to decide.

Task A: pairwise duplicate detection. Let p_1 and p_2 be two passages and let $\equiv$ denote semantic equivalence. The label is $y = 1$ if $p_1 \equiv p_2$ and $y = 0$ otherwise. The model scores the pair by $\cos(e_{p_1}, e_{p_2})$ over pooled embeddings. Task A uses no directory structure and no threshold at scoring

time, and evaluation uses the test $n \times n$ cosine matrix.

Task B: directory routing. Task B uses the same embeddings in a scholar-facing retrieval workflow: given a query passage q and the existing labelled directories, the system routes q to a directory or marks q as new. In autonomous routing, a threshold τ fit on training same-directory and different-directory pairs decides whether the query enters its nearest neighbour’s directory or the new class; in top- k ranking, each directory receives its best document match score, a pseudo-option `__NEW__` receives score τ , and the simulated expert succeeds when the labelled directory appears in the shortlist.

3.2 Dataset Construction

The corpus contains 1,705 Medieval Latin canon-law witnesses (manuscript copies of a text, each identified by its siglum: Place-Library-Shelfmark) from the Carolingian Canon Law project (CCL; Firey, 2009; Eichbauer, 2014), an open-access digital edition founded in 2009.¹ Most manuscripts are ninth-century and unedited to modern standards. Transcription is manual, from manuscript images or, for some texts, eighteenth-century printings, with one exception: handwritten text recognition has been attempted on one manuscript (Paris, BnF, lat. 1454; 142 of the 1,705 witnesses), and its transcription is still in proofreading. Before 2019 transcribers worked from microfilm and CCL contributors converted the files to TEI-P5 (the Text Encoding Initiative’s XML standard for scholarly editions). Since 2019 they transcribe in the CCL Transcription Desk beside a IIIF image of the folio (a zoomable web-served scan of the manuscript leaf); an approved proofreader then checks it against the image to the letter and adds TEI markup (CCL, 2026b). The interface credits the transcriber and proofreader of each unit (one transcribed canon or excerpt) (CCL, 2026a). We hold the witnesses as plain text derived from the CCL TEI-P5 export, with the labels as supplied.

CCL keys every unit to its original source, such as a fourth-century council or a papal decretal, and our directories reuse that key. Most of our 840 directories are named by such keys; a minority by a modern edition citation or a biblical reference. A handful carry passages whose source is unidenti-

¹Founding year, pre-2019 workflow, proofreading standard and the one HTR-transcribed manuscript: project director, personal communication, 2026.

Category	Dirs	Train	Test
Singletons (1 file)	545	272	273
Doubletons (2 files)	108	108	108
Multi-file (≥ 3)	187	467	477
Total	840	847	858

Table 1: Train and test split statistics under the v2 varied-allocation protocol. Doubleton directories are assigned as whole folders. The test set contains 535 existing files, with a same-directory partner in test, and 323 new files.

fied; each is a singleton and contributes no positive pairs. The files in a directory are the surviving witnesses to one labelled fragment, each free to vary from the others, so the benchmark asks whether an unidentified witness belongs to a known fragment. Directory sizes range from one to ten files: variants inside a directory count as positive matches, and files from different directories count as negatives.

3.3 Train/Test Split

We split the corpus into 847 training files and 858 test files with a leak-free protocol (Table 1). The 545 singleton directories are distributed at the file level. Doubleton directories are assigned as whole folders, 54 to train and 54 to test, so both files from a train doubleton contribute a positive training pair. For directories with at least three files, the train allocation varies inside each size class while keeping each class near a balanced file split, which avoids a rule where every size-three directory contributes no positive training pairs. The final split contains 565 positive training pairs and 596 positive test pairs.

All fitting uses training data only: ABTT principal components, routing thresholds, ABTT dimensionality choices, and the operational layer rule used for attribution. We train no model parameters, so the protocol uses no development split; every choice above is made on train and evaluated once on test. The one exception is the supervised fine-tuning ceiling of Tables 2 and 3, which carves its development slice out of the training files by directory, never out of test (Appendix G).

4 Methodology

Our pipeline has three stages: pool token states from one layer into a passage vector, correct the geometry, then score pairs by cosine.

4.1 Models

We evaluate six pre-trained models from three architectures. LaTa and PhilTa are T5-based models pre-trained on Latin, PhilTa with a philological

orientation, each with 12 encoder layers (Riemenschneider and Frank, 2023a), and mT5-base is a multilingual T5 encoder-decoder, also with 12 encoder layers (Xue et al., 2021). LaBSE is a language-agnostic BERT sentence encoder covering 109 languages, with 12 encoder layers (Feng et al., 2022). Qwen3-0.6B (Zhang et al., 2025) and KaLM-mini (Hu et al., 2025) are multilingual decoder-only embedding models with 28 and 24 layers.

4.2 Embedding Extraction and Layer Selection

We extract hidden-state representations from every transformer block, starting at layer 1 and skipping layer 0, the static embedding table. Token representations at each layer are pooled into one passage vector by plain mean pooling or SIF weighting (Arora et al., 2017). Appendix D compares last-token pooling, a common choice for decoder-only models, with mean pooling under ABTT.

The paper-facing pooling baseline removes only tokenizer artifacts with no lexical content, such as whitespace, padding, and tokenizer-prefix-only tokens; punctuation and numerals remain in the pooled representations. Attribution figures display original decoded tokens, even when a token contributes little.

Layer geometry and layer selection serve different roles. Label-free geometry diagnostics, including top-PC dominance, effective rank, and cosine concentration, identify the most anisotropic layers and test whether ABTT repairs them. For operational retrieval and attribution, we then apply a predeclared train-only retrieval rule: choose the earliest layer within 0.5 percentage points of the best training-set directory accuracy at rank 1 under ABTT. It selects LaTa layer 7, PhilTa layer 1, and mT5-base layer 1 for attribution, with $D = 10$ ABTT components. The most anisotropic layers are LaTa layer 8, PhilTa layer 6, and mT5-base layer 5, and we keep those as mechanism checks rather than attribution layers.

4.3 Post-Processing

ABTT. Our main post-processing method is All-but-the-Top (ABTT) (Mu and Viswanath, 2018). For each model and layer, we center the pooled training embeddings, compute the top D principal components by SVD, and remove those components from both train and test embeddings. We search $D \in \{1, 2, 3, 5, 7, 10\}$ and choose D by

training-set Task B directory accuracy at rank 1, and apply it as a retrieval-specific repair for anisotropic Latin manuscript representations.

SIF. The main comparison also includes SIF weighting (Arora et al., 2017), which divides each token weight by its estimated corpus frequency and so down-weights high-frequency tokens before the passage vector is formed. SIF and ABTT act on the same problem at different stages: SIF reweights tokens before pooling, ABTT removes dominant directions after pooling. Token probabilities for SIF come from training files only. We report four settings throughout: baseline mean pooling, SIF-only, ABTT-only, and SIF followed by ABTT.

PCA whitening (excluded). PCA whitening (Su et al., 2021) rescales principal directions rather than only removing the top components; on this corpus it degenerates the learned routing threshold, so we report its sweep only in Appendix F.

4.4 Local Attribution Methods

We evaluate local explanations for a fixed query-candidate retrieval score. The design goal is a review aid: attributions are shown as highlights connecting matching tokens across the two witnesses, so a scholar judging a proposed match can see which words the score rests on. Integrated gradients (Sundararajan et al., 2017) gives a gradient-based attribution of pairwise cosine similarity to input tokens. We compute per-sequence token attributions for each side of the pair, and the derived heatmap in Figure 5 combines token-token cosine similarity with the marginal integrated-gradient scores from the two sequences.

Our learned-mask attribution is adapted from MaRC (Brinner and Zarrieß, 2023). MaRC was proposed for classification, where a soft input mask preserves a target class score while sparsity and smoothness regularizers encourage concise rationales. We keep the per-instance mask-optimization idea and replace the classification target with pairwise retrieval cosine. The mask is optimized on one side of a query-candidate pair while the partner representation stays fixed, and swapping the two sides gives candidate-side attribution.

Let $\mathbf{E}(\mathbf{x}_q) \in \mathbb{R}^{T_q \times d_{\text{emb}}}$ denote the input-embedding lookup, $E_{>0}$ the encoder with the lookup removed, and $\mathbf{z}_c^*$ the candidate representation held fixed during optimization. We parameterize $\mathbf{m} = \sigma(\zeta)$ with free logits $\zeta \in \mathbb{R}^{T_q}$ and write $\mathcal{I}$ for content-token positions. For the ABTT

variant, define $s_{\text{ABTT}}(\mathbf{a}, \mathbf{b}) = \cos(\text{ABTT}(\mathbf{a}), \mathbf{b})$. The retrieval-adapted MaRC loss is

$$\mathcal{L}(\mathbf{m}) = (s_{\text{ABTT}}(\tilde{\mathbf{z}}_q(\mathbf{m}), \mathbf{z}_c^*) - s_{\text{ABTT}}(\mathbf{z}_q, \mathbf{z}_c^*))^2 + \lambda \sum_{i \in \mathcal{I}} \frac{m_i}{|\mathcal{I}|} + \gamma \sum_{i, i+1 \in \mathcal{I}} \frac{|m_{i+1} - m_i|}{|\mathcal{I}| - 1}, \quad (1)$$

where $\tilde{\mathbf{z}}_q(\mathbf{m}) = \text{pool}(E_{>0}(\text{diag}(\mathbf{m}) \mathbf{E}(\mathbf{x}_q)), \mathbf{a}_q)$. The baseline variant uses raw cosine in place of the ABTT-corrected cosine in Equation 1. We run 200 Adam steps per pair, initialize $\zeta_0 = \log 9$, and freeze logits outside $\mathcal{I}$ at $-\infty$.

4.5 Evaluation Metrics

Task A. Both Task A metrics come from the test $n \times n$ pairwise cosine matrix. AUROC treats same-directory test pairs as positives and different-directory pairs as negatives. Cosine gap is the mean same-directory cosine minus the mean different-directory cosine.

Task B. For autonomous routing, we report overall assignment accuracy, existing accuracy, and new accuracy. Assignment accuracy scores only the existing-versus-new decision: a file with a partner in test is correct when its maximum cosine to another test file is at least τ , and a file without one is correct when that maximum falls below τ . It never checks which directory the file would enter, so a degenerate threshold that routes everything as existing already attains the 62.4 percent class prior; read it next to directory accuracy at rank 1, which does require the correct directory. For top- k ranking, cumulative top- K accuracy is the fraction of test queries whose labelled directory appears among the top K options.

Attribution. For attribution, we adapt ERASER-style sufficiency and comprehensiveness metrics (DeYoung et al., 2020) to retrieval cosine. Sufficiency keeps the top-ranked query tokens and measures how much of the full pair cosine remains, and comprehensiveness removes those tokens and measures the score drop. MinFrac reports the smallest retained query-token fraction needed to recover a fixed share of the full cosine. We also report ρ_{LOO} , the Spearman correlation between attribution magnitude and the leave-one-out drop in the same retrieval score. Every curve metric ranks tokens by attribution magnitude $|a|$ with ties broken by token position; Appendix H states both conventions and their consequences.

5 Results

5.1 Task A: Pairwise Duplicate Detection

In Table 2, baseline quality is uneven, with AUROC from 0.838 for mT5-base to 0.972 for KaLM-mini, and ABTT removes most of that unevenness: every model lands between 0.971 and 0.987, and the cosine gap rises from a baseline range of 0.024 to 0.237 to a range of 0.525 to 0.611. The gain is largest where the baseline is weakest: mT5-base gains 0.137 AUROC, LaTa 0.034, and KaLM-mini 0.009.

The cosine gap column explains why a strong AUROC is not enough: KaLM-mini has the best baseline AUROC, 0.972, but a gap of only 0.056, and Qwen3-0.6B ranks nearly as well, 0.966, with the smallest gap of all, 0.024, so both leave almost no margin for a threshold. ABTT raises every model’s gap by at least 0.28, the margin a single learned threshold needs in Task B.

SIF as a frequency-based complement. Tables 2 and 3 show how far SIF, the frequency side of the same dominant direction, gets. On Task B assignment accuracy, SIF-only adds 9.4 points for LaTa, 8.2 for PhilTa, and 7.3 for mT5-base, the three models whose retrieval collapses; it adds 0.1 point for LaBSE and subtracts 1.9 for KaLM-mini and 6.3 for Qwen3-0.6B. SIF-only never matches ABTT-only on either task for any model, and adding SIF before ABTT moves best-layer assignment accuracy by at most 1.8 points either way. SIF is a partial account of the anisotropy, not a competing repair; the full sweeps are in Appendix F.

5.2 Layerwise Geometry Diagnosis

The layerwise profile separates the two effects: the collapse belongs to the T5 encoders. Baseline AUROC bottoms out at 0.496 for LaTa at layer 6, which is chance, at 0.538 for PhilTa at layer 10, and at 0.654 for mT5-base at layer 5, while ABTT at those layers gives 0.964, 0.982, and 0.978. For the other three, baseline AUROC rises with depth to a final-layer AUROC of 0.956 for LaBSE, 0.958 for Qwen3-0.6B, and 0.956 for KaLM-mini, with layer minima of 0.806, 0.858, and 0.861. Figure 1 plots the cosine gap across depth and Figure 2 (Appendix A) plots AUROC over the same panels.

Anisotropy, in contrast, is present in all six. At each model’s most anisotropic layer, the layer with the largest top-PC variance share, the mean pair-

wise cosine on the test split is 0.230 for LaTa, 0.584 for PhilTa, 0.314 for mT5-base, 0.878 for LaBSE, 0.953 for Qwen3-0.6B, and 0.887 for KaLM-mini. ABTT with $D = 10$ drives all six to within 0.002 of zero and raises effective rank from as low as 1.00 to between 147.50 and 176.36 (Appendix A). The T5 encoders concentrate variance in one direction, with top-PC shares of 0.956, 0.862, and 1.000, so the cone is low-rank and retrieval dies with it. The non-T5 models hold top-PC shares of 0.506, 0.167, and 0.396 but still carry a large common component, so passages sit close together without losing their relative order. ABTT removes that common component in both cases, which is why the repair generalizes even though the symptom does not.

At PhilTa’s collapsed retrieval layer, equivalent and non-equivalent cosines overlap heavily before ABTT and separate after it (Figure 3, Appendix B).

In t-SNE (van der Maaten and Hinton, 2008) and UMAP (McInnes et al., 2018) projections of all six models (Appendix J), ABTT-only with $D = 10$ forms tighter same-directory groups than baseline mean pooling, with winnable-witness silhouette scores rising by 0.07 to 0.72 on the three T5 encoders.

5.3 Task B: Directory Routing

In Table 3, directory accuracy at rank 1 spans 39.3 points under the baseline, from 46.6 for mT5-base to 85.9 for KaLM-mini, so the model choice dominates. ABTT compresses that to 3.3 points, from 86.1 to 89.4. Assignment accuracy tracks it, moving from a 40.3-point spread to the range 88.5 to 91.7, with every ABTT cell above every baseline cell.

Tables 13 and 16 in Appendix F repeat the top-1 comparison under five query and reference reseedings at each method’s own train-selected layer (the Base and SIF+ABTT subscripts of Table 3), so the compression is not an artifact of one draw. The baseline spread of 36.6 points, from 50.7 for mT5-base to 87.3 for KaLM-mini, falls to 1.7 points under SIF+ABTT, from 88.9 for LaTa to 90.6 for LaBSE, with standard deviations of at most 1.0. Table 13 gives the shortlist view (per-layer: Appendix C): every model exceeds 95% by Top-2 and gains little after Top-3.

Lexical baselines set the operating point. Witnesses to one fragment are hand copies of one exemplar and differ mostly in spelling, so surface overlap is strong here. Character 3–5-gram TF-IDF

Model	Task A AUROC				Task A cosine gap			
	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	0.938 ₁₂	0.948 ₁	0.971 ₁₂	0.966 ₇	0.237 ₁₂	0.366 ₁	0.525 ₁₂	0.580 ₇
PhilTa	0.939 ₁	0.955 ₁	0.983 ₉	0.982 ₈	0.231 ₁	0.319 ₁	0.540 ₉	0.573 ₈
mT5-base	0.838 ₁₂	0.845 ₁	0.975 ₂	0.974 ₁	0.081 ₁₂	0.036 ₁	0.536 ₂	0.519 ₁
LaBSE	0.956 ₁₂	0.956 ₁₂	0.987 ₁₁	0.976 ₁₀	0.183 ₁₂	0.184 ₁₂	0.611 ₁₁	0.580 ₁₀
Qwen3-0.6B	0.966 ₂₆	0.942 ₂₂	0.973 ₂	0.972 ₂	0.024 ₂₆	0.053 ₂₂	0.553 ₂	0.540 ₂
KaLM-mini	0.972 ₂₃	0.964 ₂₃	0.981 ₁	0.981 ₂	0.056 ₂₃	0.056 ₂₃	0.568 ₁	0.535 ₂
LaTa (fine-tuned)	0.984 ₁₂	–	0.970 ₁₂	–	0.387 ₁₂	–	0.548 ₁₂	–
BM25 (word)			0.974				0.243	
TF-IDF char 3–5			0.987				0.555	
Levenshtein			0.952				0.362	

Table 2: Task A pairwise duplicate detection for all six models under four post-processing settings. Each cell is a test-set score at the layer chosen by training-set AUROC, given as the subscript; ABTT uses D tuned on train. Baseline AUROC spans 0.838 to 0.972, and ABTT lifts every model into a 0.971 to 0.987 band. Cosine gap is defined in Figure 1. Per-layer grids: Appendix Tables 6, 10, and 14. Below the rule, reference systems on the same split with the same evaluation code: LaTa fine-tuned contrastively on 499 of the 565 positive train pairs (a 28-directory dev curve takes the rest), a ceiling at this training budget, and three lexical baselines fitted on train files (both setups: Appendix G). TF-IDF char 3–5 is level with the best ABTT AUROC (0.987 against 0.987), and ABTT moves the fine-tuned encoder’s AUROC from 0.984 to 0.970 while moving its gap from 0.387 to 0.548. Surface overlap is the practitioner’s operating point on this corpus, and the embedding rows diagnose representation geometry rather than beat it.

Model	Assignment accuracy				Directory accuracy @1			
	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	73.8 ₁	83.2 ₁	88.5 ₈	90.2 ₁₂	72.1 ₁	81.8 ₁	86.1 ₈	88.2 ₁₂
PhilTa	70.9 ₁	79.0 ₁	91.1 ₁	91.4 ₁	69.3 ₁	77.7 ₁	88.3 ₁	88.9 ₁
mT5-base	47.2 ₁₂	54.5 ₁	90.3 ₁	90.3 ₂	46.6 ₁₂	54.1 ₁	88.5 ₁	87.5 ₂
LaBSE	83.3 ₁₁	83.4 ₁₂	90.8 ₁₁	89.6 ₁₁	81.5 ₁₁	81.9 ₁₂	88.5 ₁₁	86.9 ₁₁
Qwen3-0.6B	82.3 ₂₈	76.0 ₂₇	91.5 ₅	90.8 ₅	80.3 ₂₈	74.2 ₂₇	89.4 ₅	88.6 ₅
KaLM-mini	87.5 ₂₃	85.7 ₂₂	91.7 ₃	90.8 ₁	85.9 ₂₃	84.0 ₂₂	89.4 ₃	88.0 ₁
LaTa (fine-tuned)	83.4 ₁₂	–	87.8 ₁₂	–	81.6 ₁₂	–	85.2 ₁₂	–
BM25 (word)			62.5				56.3	
TF-IDF char 3–5			91.8				89.9	
Levenshtein			87.8				86.2	

Table 3: Task B autonomous routing for all six models under four post-processing settings, in percent. Each cell is a test-set score at the layer chosen by training-set directory accuracy at rank 1, given as the subscript. Assignment accuracy scores the existing-versus-new decision alone, comparing each file’s maximum cosine against the train-fit threshold τ ; a degenerate threshold that routes everything as existing already attains the 62.4 percent class prior. Directory accuracy at rank 1 requires the correct directory, so read the two columns together. Per-layer grids: Appendix Tables 7, 11, and 15. Below the rule, reference systems on the same split with the same evaluation code: LaTa fine-tuned contrastively on 499 of the 565 positive train pairs (a 28-directory dev curve takes the rest), a ceiling at this training budget, and three lexical baselines fitted on train files (both setups: Appendix G). TF-IDF char 3–5 is level with the best ABTT cell (91.8 against 91.7 assignment accuracy, 89.9 against 89.4 directory accuracy at rank 1), and the fine-tuned encoder with ABTT (87.8 and 85.2) is below every zero-shot ABTT cell (88.5 to 91.7 and 86.1 to 89.4). Surface overlap is the practitioner’s operating point on this corpus, and the embedding rows diagnose representation geometry rather than beat it.

(settings in Appendix G) matches the best ABTT cell on Task A (AUROC 0.987 for both) and is level with it on Task B (91.8 against 91.7 assignment accuracy, 89.9 against 89.4 directory accuracy at rank 1, one draw against seed standard deviations of up to 1.0). Stratified by TF-IDF overlap, the 596 positive test pairs show no regime where an embedding ranks better: TF-IDF is never worse in any tercile, leads LaBSE+ABTT corpus-wide

(recall at rank 1 of 0.878 against 0.854, McNemar $p = 0.001$), leads only marginally on low-overlap pairs ($p = 0.023$ and 0.033, uncorrected over twelve comparisons per stratum), and is at parity with LaBSE+ABTT’s one routing edge there ($p = 0.33$). Word-level BM25 ranks well (AUROC 0.974) but routes no better than the 62.4 percent class prior, so the character baseline is the comparison that matters.

Model	Method	ρ_{LOO}		DeLAUC gap	
		base	ABTT	base	ABTT
LaTa	IG	0.013	0.298	0.504	0.178
	MaRC	0.070	0.397	0.244	0.287
PhilTa	IG	0.144	0.577	0.806	0.420
	MaRC	0.179	0.367	0.758	0.353
mT5-base	IG	0.138	0.626	0.394	0.548
	MaRC	0.272	0.415	0.075	0.431

Table 4: Attribution faithfulness at the predeclared operational layers, 200 positive pairs per model, for integrated gradients (IG) and retrieval-adapted MaRC. ρ_{LOO} correlates attribution magnitude with the leave-one-out change in the cosine. DeLAUC gap is the deletion-curve area in attribution order minus the area under a random order, whose reference runs from 0.671 to 0.928 here, so zero is chance. Higher is better in both; boldface marks the better variant. ABTT wins 6/6 and 3/6, and the LaTa MaRC DeLAUC win is a tie at about 1.2 standard errors. Ratio metrics are undefined below a full-query cosine of 0.05, so the DeLAUC columns average 191 to 195 ABTT pairs against 200 baseline pairs. Cross-variant comparisons are descriptive. Secondary metrics: Table 19.

A parameter-free repair matches contrastive fine-tuning. Fine-tuning LaTa on the positive train pairs (499 after a directory-level dev carve; setup in Appendix G) gives the best uncorrected Task A row (AUROC 0.984) and 83.4 assignment accuracy; with ABTT on top it reaches 87.8 assignment accuracy and 85.2 directory accuracy at rank 1, below every zero-shot ABTT cell (88.5 to 91.7 and 86.1 to 89.4). ABTT also lowers that row’s AUROC from 0.984 to 0.970 while widening its cosine gap from 0.387 to 0.548, consistent with fine-tuning having already removed part of the common component: the repair is not free on every row, and it matches what the available supervision buys at this training budget.

5.4 Attribution at Retrieval-Selected Layers

Table 4 compares baseline and ABTT variants of integrated gradients and retrieval-adapted MaRC on 200 positive query-candidate pairs per model for LaTa, PhilTa, and mT5-base. ABTT repairs the scores being explained at these layers: LaTa layer 7 moves from AUROC 0.498 to 0.962 and directory accuracy 0.302 to 0.868, PhilTa layer 1 from 0.939 to 0.977 and 0.693 to 0.883, and mT5-base layer 1 from 0.822 to 0.979 and 0.451 to 0.885.

ABTT improves rank faithfulness in all six model-view cells (ρ_{LOO} 6/6, with tie-corrected Kendall τ_b agreeing in all six), but chance-corrected deletion faithfulness improves in only three of six,

and one of those three wins is within 1.2 standard errors of zero. The sufficiency-side metrics fail our shuffled-attribution control in two baseline cells and stay in the appendix, so we claim better-ranked rationales, not a gain on every axis.

Masking acts on cached layer- L hidden states, so every faithfulness statement is about the pooled read-out at that layer, not the encoder’s input-to-output map. Appendix H carries τ_b , chance-corrected insertion faithfulness, the ERASER-style rationale metrics and their threshold sweeps (Table 19, Figure 4); Appendix I shows one PhilTa retrieval case, with its integrated-gradient pair matrix before and after ABTT.

6 Discussion

Six models across three architectures narrow what we can claim. Anisotropy is universal on this corpus and so is the repair, but the mid-depth retrieval collapse is not. Layerwise profiles place the mid-depth anisotropy peak in decoder-only models and do not separate encoder-decoder encoders (Razhigaev et al., 2024; Godey et al., 2024), so, to our knowledge, this is the first report of the collapse for T5 encoders, and we do not read it as a property of the architecture. A single-architecture study could report that collapse as the general finding.

With a T5 encoder the middle layers are unusable without correction, so layer choice matters. With a non-T5 model the final layers already work, and ABTT still adds 3.5 to 9.1 points of directory accuracy at rank 1.

Geometry diagnoses the failure mode but does not select layers: we use it to locate where representations collapse and recover, and a train-only task signal to pick retrieval layers. The selected layers are not the most anisotropic ones. The attribution results at those locked layers need a narrow reading: ABTT changes the cosine being explained, so only the leave-one-out rank gain is stable across all six cells.

The repair also changes what a scholar sees. Under SIF+ABTT every model puts the correct directory in a two-item shortlist for more than 95% of test queries, with token attributions ranking the words the score rests on. The existing-versus-new call is still wrong for roughly one file in ten, so the open-set decision stays with the scholar. All six encoders share the geometry. Only the T5 encoders let it cost them retrieval, and one SVD per layer, fit on 847 training files, removes that cost.

Limitations

The study covers one domain, medieval canon law, and one language, Latin. The 1,705-witness corpus is large for this field, but small by current NLP standards. We test six models; others may behave differently. The mid-depth retrieval collapse we document holds for the three T5 encoders in our set, and we cannot separate architecture, pre-training objective, and checkpoint as its cause. ABTT also needs enough text to estimate principal components, so smaller archives need separate study.

Task B simulates a human-in-the-loop workflow by counting a query as successful when its labelled directory appears in the shortlist. That measures ranking quality, not the full judgment load a working scholar carries. Our attribution analysis is also local to fixed query-candidate scores. It does not explain model behavior globally, and it does not replace expert reading of retrieved passages. The consistent attribution effect is improved leave-one-out rank faithfulness; other rationale metrics remain threshold-dependent and should not be reported as a broad attribution-quality gain.

References

- Mira Ait-Saada and Mohamed Nadif. 2023. [Is anisotropy truly harmful? A case study on text clustering](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 1194–1203. Association for Computational Linguistics.
- Avishek Anand, Lijun Lyu, Maximilian Idahl, Yumeng Wang, Jonas Wallat, and Zijian Zhang. 2022. [Explorable information retrieval: A survey](#). *arXiv preprint arXiv:2211.02405*.
- Sanjeev Arora, Yingyu Liang, and Tengyu Ma. 2017. [A simple but tough-to-beat baseline for sentence embeddings](#). In *Proceedings of the 5th International Conference on Learning Representations (ICLR)*.
- Pepa Atanasova, Jakob Grue Simonsen, Christina Lioma, and Isabelle Augenstein. 2020. [A diagnostic study of explainability techniques for text classification](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3256–3274. Association for Computational Linguistics.
- David Bamman and Patrick J. Burns. 2020. [Latin BERT: A contextual language model for classical philology](#). *arXiv preprint arXiv:2009.10053*.
- Marc Brinner and Sina Zarrieß. 2023. [Model interpretability and rationale extraction by input mask optimization](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13722–13744. Association for Computational Linguistics.

- Patrick J. Burns, James A. Brofos, Kyle Li, Pramit Chaudhuri, and Joseph P. Dexter. 2021. [Profiling of intertextuality in Latin literature using word embeddings](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4900–4907. Association for Computational Linguistics.
- CCL. 2026a. [About the Project](#). Carolingian Canon Law Project home page, University of Kentucky. Accessed 2026-09-14.
- CCL. 2026b. [Submission Desks](#). Carolingian Canon Law Project documentation page, University of Kentucky. Accessed 2026-09-14.
- Neil Coffee, Jean-Pierre Koenig, Shakthi Poornima, Roelant Ossewaarde, Christopher Forstall, and Sarah Jacobson. 2012. [Intertextuality in the digital age](#). *Transactions of the American Philological Association*, 142(2):383–422.
- Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C. Wallace. 2020. [ERASER: A benchmark to evaluate rationalized NLP models](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4443–4458, Online. Association for Computational Linguistics.
- Melodie H. Eichbauer. 2014. [Review of the Carolingian Canon Law Project, directed by Abigail Firey](#). *Digital Philology: A Journal of Medieval Cultures*, 3(1):165–168.
- Kawin Ethayarajh. 2019. [How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 55–65. Association for Computational Linguistics.
- Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. 2022. [Language-agnostic BERT sentence embedding](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 878–891. Association for Computational Linguistics.
- Abigail Firey. 2009. [The Carolingian Canon Law Project](#). Collaboratory for Research in Computing for the Humanities, University of Kentucky. Open-access digital edition, ongoing. Accessed 2026-09-14.
- Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, and Tie-Yan Liu. 2019. [Representation degeneration problem in training natural language generation models](#). In *Proceedings of the 7th International Conference on Learning Representations (ICLR)*.

729	Nathan Godey, Éric de la Clergerie, and Benoît Sagot.	allusive text reuse . In <i>Proceedings of the 3rd Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature</i> , pages 104–114. Association for Computational Linguistics.	786
730	2024. Anisotropy is inherent to self-attention in trans-		787
731	formers . In <i>Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages		788
732	35–48. Association for Computational Linguistics.		789
733			790
734			
735	Xinshuo Hu, Zifei Shan, Xinping Zhao, Zetian Sun,	Leland McInnes, John Healy, and James Melville. 2018.	791
736	Zhenyu Liu, Dongfang Li, Shaolin Ye, Xinyuan Wei,	UMAP: Uniform manifold approximation and pro-	792
737	Qian Chen, Baotian Hu, Haofen Wang, Jun Yu, and	jection for dimension reduction . <i>arXiv preprint</i>	793
738	Min Zhang. 2025. KaLM-embedding: Superior train-	<i>arXiv:1802.03426</i> .	794
739	ing data brings a stronger embedding model . <i>arXiv</i>		
740	<i>preprint arXiv:2501.01028</i> .	Jiaqi Mu and Pramod Viswanath. 2018. All-but-the-	795
741		Top: Simple and effective postprocessing for word	796
742	Alon Jacovi and Yoav Goldberg. 2020. Towards faith-	representations . In <i>Proceedings of the 6th Inter-</i>	797
743	fully interpretable NLP systems: How should we	<i>national Conference on Learning Representations</i>	798
744	define and evaluate faithfulness? In <i>Proceedings</i>	<i>(ICLR)</i> .	799
745	<i>of the 58th Annual Meeting of the Association for</i>		
746	<i>Computational Linguistics</i> , pages 4198–4205. Asso-	Niklas Muennighoff, Nouamane Tazi, Loic Magne, and	800
747	ciation for Computational Linguistics.	Nils Reimers. 2023. MTEB: Massive text embedding	801
748		benchmark . In <i>Proceedings of the 17th Conference of</i>	802
749	Sarthak Jain and Byron C. Wallace. 2019. Attention	<i>the European Chapter of the Association for Compu-</i>	803
750	is not explanation . In <i>Proceedings of the 2019 Con-</i>	<i>tational Linguistics</i> , pages 2014–2037. Association	804
751	<i>ference of the North American Chapter of the Asso-</i>	for Computational Linguistics.	805
752	<i>ciation for Computational Linguistics: Human Lan-</i>		
753	<i>guage Technologies, Volume 1 (Long and Short Pa-</i>	Jianmo Ni, Gustavo Hernandez Abrego, Noah Constant,	806
754	<i>pers)</i> , pages 3543–3556. Association for Computa-	Ji Ma, Keith Hall, Daniel Cer, and Yinfei Yang. 2022.	807
755	tional Linguistics.	Sentence-T5: Scalable sentence encoders from pre-	808
756		trained text-to-text models . In <i>Findings of the As-</i>	809
757	Olga Kovaleva, Saurabh Kulshreshtha, Anna Rogers,	<i>sociation for Computational Linguistics: ACL 2022</i> ,	810
758	and Anna Rumshisky. 2021. BERT busters: Outlier	pages 1864–1874. Association for Computational	811
759	dimensions that disrupt transformers . In <i>Findings of</i>	Linguistics.	812
760	<i>the Association for Computational Linguistics: ACL-</i>		
761	<i>IJCNLP 2021</i> , pages 3392–3405. Association for	Sara Rajae and Mohammad Taher Pilehvar. 2022. An	813
762	Computational Linguistics.	isotropy analysis in the multilingual BERT embed-	814
763		ding space . In <i>Findings of the Association for Com-</i>	815
764	Nelson F. Liu, Matt Gardner, Yonatan Belinkov,	<i>putational Linguistics: ACL 2022</i> , pages 1309–1316.	816
765	Matthew E. Peters, and Noah A. Smith. 2019. Lin-	Association for Computational Linguistics.	817
766	guistic knowledge and transferability of contextual		
767	representations . In <i>Proceedings of the 2019 Confer-</i>	Anton Razzhigaev, Matvey Mikhalechuk, Elizaveta Gon-	818
768	<i>ence of the North American Chapter of the Associ-</i>	charova, Ivan Oseledets, Denis Dimitrov, and Andrey	819
769	<i>ation for Computational Linguistics: Human Lan-</i>	Kuznetsov. 2024. The shape of learning: Anisotropy	820
770	<i>guage Technologies, Volume 1 (Long and Short Pa-</i>	and intrinsic dimensions in transformer-based mod-	821
771	<i>pers)</i> , pages 1073–1094. Association for Computa-	els . In <i>Findings of the Association for Computational</i>	822
772	tional Linguistics.	<i>Linguistics: EACL 2024</i> , pages 868–874. Association	823
773		for Computational Linguistics.	824
774	Yifan Liu, Gelila Tilahun, Xinxiang Gao, Qianfeng Wen,		
775	and Michael Gervers. 2025. A comparative study	Frederick Riemenschneider and Anette Frank. 2023a.	825
776	of static and contextual embeddings for analyzing	Exploring large language models for classical philol-	826
777	semantic changes in medieval Latin charters . In <i>Pro-</i>	ogy . In <i>Proceedings of the 61st Annual Meeting of</i>	827
778	<i>ceedings of the First Workshop on Language Models</i>	<i>the Association for Computational Linguistics (Vol-</i>	828
779	<i>for Low-Resource Languages</i> , pages 182–192, Abu	<i>ume 1: Long Papers)</i> , pages 15181–15199. Associa-	829
780	Dhabi, United Arab Emirates. Association for Com-	tion for Computational Linguistics.	830
781	putational Linguistics.		
782	Anemily Machina and Robert Mercer. 2024. Anisotropy	Frederick Riemenschneider and Anette Frank. 2023b.	831
783	is not inherent to transformers . In <i>Proceedings of</i>	Graecia capta ferum victorem cepit. Detecting Latin	832
784	<i>the 2024 Conference of the North American Chap-</i>	allusions to Ancient Greek literature . In <i>Proceedings</i>	833
785	<i>ter of the Association for Computational Linguistics:</i>	<i>of the Ancient Language Processing Workshop</i> , pages	834
	<i>Human Language Technologies (Volume 1: Long</i>	30–38, Varna, Bulgaria. INCOMA Ltd., Shoumen,	835
	<i>Papers)</i> , pages 4892–4907. Association for Compu-	Bulgaria.	836
	tational Linguistics.		
	Enrique Manjavacas, Brian Long, and Mike Kestemont.	Oscar Skean, Md Rifat Arefin, Dan Zhao, Niket Nikul	837
	2019. On the feasibility of automated detection of	Patel, Jalal Naghiyev, Yann LeCun, and Ravid	838
		Shwartz-Ziv. 2025. Layer by layer: Uncovering hid-	839
		den representations in language models . In <i>Proceed-</i>	840
		<i>ings of the 42nd International Conference on Ma-</i>	841
		<i>chine Learning (ICML)</i> , volume 267 of <i>Proceedings</i>	842

of Machine Learning Research, pages 55854–55875. PMLR.

David A. Smith, Ryan Cordell, and Elizabeth Maddock Dillon. 2013. [Infectious texts: Modeling text reuse in nineteenth-century newspapers](#). In *2013 IEEE International Conference on Big Data*, pages 86–94. IEEE.

Rachele Sprugnoli, Marco Passarotti, Flavio Massimiliano Cecchini, and Matteo Pellegrini. 2020. [Overview of the EvaLatin 2020 evaluation campaign](#). In *Proceedings of LT4HALA 2020 - 1st Workshop on Language Technologies for Historical and Ancient Languages*, pages 105–110. European Language Resources Association (ELRA).

Jianlin Su, Jiarun Cao, Weijie Liu, and Yangyiwen Ou. 2021. [Whitening sentence representations for better semantics and faster retrieval](#). *arXiv preprint arXiv:2103.15316*.

Mukund Sundararajan, Ankur Taly, and Qiqi Yan. 2017. [Axiomatic attribution for deep networks](#). In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *Proceedings of Machine Learning Research*, pages 3319–3328. PMLR.

Ian Tenney, Dipanjan Das, and Ellie Pavlick. 2019. [BERT rediscovers the classical NLP pipeline](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601. Association for Computational Linguistics.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. [BEIR: A heterogenous benchmark for zero-shot evaluation of information retrieval models](#). In *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

William Timkey and Marten van Schijndel. 2021. [All bark and no bite: Rogue dimensions in transformer language models obscure representational quality](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 4527–4546. Association for Computational Linguistics.

Laurens van der Maaten and Geoffrey Hinton. 2008. [Visualizing data using t-SNE](#). *Journal of Machine Learning Research*, 9(86):2579–2605.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. [mT5: A massively multilingual pre-trained text-to-text transformer](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 483–498. Association for Computational Linguistics.

Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren

Zhou. 2025. [Qwen3 embedding: Advancing text embedding and reranking through foundation models](#). *arXiv preprint arXiv:2506.05176*.

A Layer Diagnostics and Attribution Layer Selection

This appendix separates the most anisotropic layers, where geometry collapses, from the operational attribution layers, which are selected by training-set retrieval accuracy. Table 5 lists both for the three T5 encoders, together with the top-PC share and effective rank before and after ABTT. Figure 2 completes the layerwise view of Figure 1 by plotting Task A AUROC against depth for all six models; it is the view that shows the mT5-base collapse, a ranking failure that the cosine gap does not register because that gap rises in the middle layers.

Across the main-model layer grid, PC1 dominance and low effective rank are strongly associated with ABTT gains over baseline. These associations explain why ABTT helps, but they do not replace the train-only retrieval rule used for layer selection.

B Score Distributions at the Collapsed Retrieval Layer

Figure 3 restates the collapse and the repair as score distributions rather than as summary metrics, for the model whose collapse is easiest to read.

C Expanded Per-Layer Results for T5 Encoders

The main text compresses the headline retrieval results into Tables 2 and 3, which report one train-selected layer per model and method. This appendix provides the expanded per-layer Task A duplicate-detection, Task B autonomous-routing, and Task B top- k ranking tables for LaTa, PhilTa, and mT5-base; Appendix E gives the same view for the other three models.

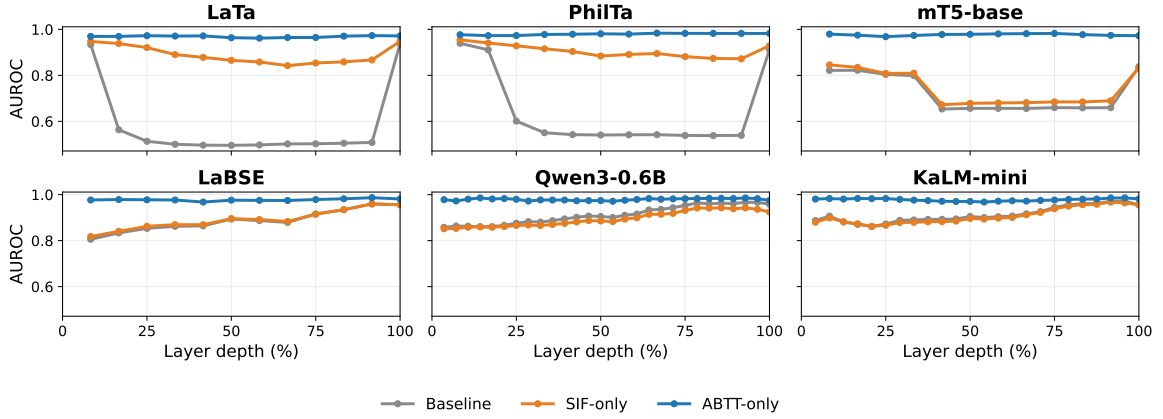


Figure 2: Task A AUROC against normalized layer depth, same panels and same series as Figure 1: gray is baseline mean pooling, blue is ABTT-only with D tuned on train, orange is SIF-only. In the top row, LaTa and PhilTa fall to chance in their middle layers and mT5-base loses more than half of its above-chance separation; for mT5-base this panel is the only view of the collapse, since its baseline cosine gap in Figure 1 rises over the same layers. The three non-T5 models in the bottom row improve close to monotonically with depth and never fall below 0.80. ABTT holds every model near its ceiling at every depth, so the repair does not depend on the collapse being present.

Model	Operational layer	Most anisotropic layer	PC1 before	PC1 after	Eff. rank before→after
LaTa	7	8	0.956	0.036	1.34→168.40
PhilTa	1	6	0.862	0.039	1.83→155.02
mT5-base	1	5	1.000	0.044	1.00→151.74

Table 5: Most anisotropic layers, by top-PC variance share, and operational attribution layers for the three T5 encoders. PC1 and effective-rank values compare baseline geometry with ABTT-D10 at the most anisotropic layer. Operational layers come from the train-only retrieval rule.

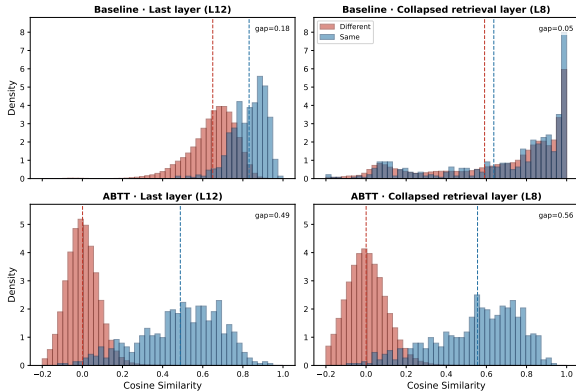


Figure 3: Cosine similarity distributions for PhilTa at the last layer and at its collapsed retrieval layer, before and after ABTT. Equivalent pairs share a directory (Same in the legend); non-equivalent pairs do not (Different). The collapsed retrieval layer is layer 8, the layer with the highest training-set AUROC within 30 to 70 percent of encoder depth (layers 4 to 8), so the panel understates the collapse rather than selecting its worst case; the same train-only rule that fixes every other choice in the paper picks it, and its test AUROC is 0.542, close to chance. Two other PhilTa layers carry other meanings and are not shown here: the most anisotropic layer under the top-PC rule of Appendix A is layer 6, and the lowest baseline test AUROC, 0.538, falls at layer 10. Before ABTT the two distributions sit almost on top of each other, and ABTT separates them.

Task A: Pairwise Duplicate Detection (main)					
Model	Layer	AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
LaTa	1	0.934	0.970	0.286	0.566
	2	0.563	0.969	0.019	0.554
	3	0.513	0.973	-0.048	0.555
	4	0.500	0.971	-0.067	0.565
	5	0.496	0.972	-0.069	0.572
	6	0.496	0.964	-0.069	0.569
	7	0.498	0.962	-0.072	0.564
	8	0.502	0.965	-0.074	0.558
	9	0.503	0.965	-0.073	0.557
	10	0.505	0.971	-0.072	0.551
	11	0.508	0.973	-0.067	0.550
	12	0.938	0.971	0.237	0.525
PhilTa	1	0.939	0.977	0.231	0.574
	2	0.911	0.973	0.161	0.553
	3	0.601	0.973	0.085	0.549
	4	0.551	0.978	0.063	0.551
	5	0.542	0.979	0.053	0.551
	6	0.541	0.981	0.047	0.551
	7	0.542	0.979	0.046	0.554
	8	0.542	0.984	0.046	0.555
	9	0.539	0.983	0.041	0.540
	10	0.538	0.982	0.040	0.531
	11	0.539	0.982	0.041	0.511
	12	0.911	0.982	0.181	0.487
mT5-base	1	0.822	0.979	0.027	0.550
	2	0.822	0.975	0.026	0.536
	3	0.804	0.968	0.025	0.517
	4	0.799	0.974	0.040	0.502
	5	0.654	0.978	0.219	0.482
	6	0.657	0.978	0.238	0.483
	7	0.656	0.980	0.244	0.482
	8	0.656	0.982	0.249	0.471
	9	0.660	0.982	0.235	0.475
	10	0.659	0.977	0.226	0.459
	11	0.659	0.974	0.213	0.449
	12	0.838	0.972	0.081	0.442

Table 6: Per-layer Task A pairwise duplicate-detection metrics for the three T5 encoders (LaTa, PhilTa, mT5-base). baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. Both AUROC and cosine gap are computed over the test $n \times n$ cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the best layer per model, selected by ABTT AUROC.

Task B: Autonomous Routing (main, file-level, τ -thresholded)							
Model	Layer	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
LaTa	1	0.630	0.910	0.916	0.907	0.738	0.909
	2	0.153	0.905	0.913	0.892	0.439	0.900
	3	0.256	0.882	0.842	0.950	0.477	0.908
	4	0.389	0.893	0.718	0.920	0.513	0.903
	5	0.374	0.892	0.715	0.913	0.502	0.900
	6	0.378	0.884	0.712	0.913	0.503	0.895
	7	0.372	0.875	0.715	0.916	0.501	0.890
	8	0.379	0.884	0.706	0.885	0.502	0.885
	9	0.366	0.886	0.724	0.873	0.501	0.881
	10	0.344	0.873	0.752	0.882	0.498	0.876
	11	0.310	0.867	0.786	0.901	0.490	0.880
	12	0.576	0.890	0.932	0.833	0.710	0.868
PhilTa	1	0.591	0.918	0.904	0.901	0.709	0.911
	2	0.479	0.895	0.867	0.885	0.625	0.892
	3	0.193	0.897	0.889	0.879	0.455	0.890
	4	0.148	0.886	0.923	0.873	0.439	0.881
	5	0.252	0.879	0.796	0.873	0.457	0.876
	6	0.338	0.905	0.700	0.817	0.474	0.872
	7	0.301	0.901	0.746	0.830	0.469	0.874
	8	0.262	0.875	0.808	0.885	0.467	0.879
	9	0.303	0.890	0.762	0.870	0.476	0.882
	10	0.299	0.908	0.780	0.793	0.480	0.865
	11	0.256	0.869	0.824	0.836	0.470	0.857
	12	0.338	0.880	0.898	0.876	0.549	0.879
mT5-base	1	0.135	0.877	0.981	0.947	0.453	0.903
	2	0.110	0.890	0.975	0.910	0.436	0.897
	3	0.161	0.794	0.926	0.941	0.449	0.850
	4	0.269	0.787	0.858	0.882	0.491	0.823
	5	0.422	0.746	0.659	0.864	0.512	0.790
	6	0.636	0.766	0.489	0.876	0.580	0.808
	7	0.594	0.793	0.517	0.851	0.565	0.815
	8	0.557	0.809	0.560	0.842	0.558	0.822
	9	0.364	0.841	0.746	0.814	0.508	0.831
	10	0.314	0.794	0.783	0.876	0.491	0.825
	11	0.297	0.763	0.780	0.854	0.479	0.797
	12	0.198	0.834	0.926	0.799	0.472	0.821

Table 7: Per-layer Task B autonomous routing accuracy for the three T5 encoders. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. For every test file we compute $s_i = \max_{j \neq i} \cos(e_i, e_j)$ against the rest of the test set and compare to the learned threshold τ (fit on train via F_1 -optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if $s_i \geq \tau$, else new); only files predicted existing are then routed to their top-1 neighbour’s directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the best layer per model, selected by ABTT overall assignment accuracy.

Task B: Top-K Ranking (main, single-seed v2)							
Model	Layer	Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
LaTa	1	0.721	0.880	0.630	0.910	0.916	0.907
	2	0.383	0.869	0.153	0.905	0.913	0.892
	3	0.350	0.882	0.256	0.882	0.842	0.950
	4	0.303	0.881	0.389	0.893	0.718	0.920
	5	0.297	0.875	0.374	0.892	0.715	0.913
	6	0.297	0.874	0.378	0.884	0.712	0.913
	7	0.302	0.868	0.372	0.875	0.715	0.916
	8	0.302	0.861	0.379	0.884	0.706	0.885
	9	0.309	0.858	0.366	0.886	0.724	0.873
	10	0.317	0.852	0.344	0.873	0.752	0.882
	11	0.331	0.857	0.310	0.867	0.786	0.901
	12	0.689	0.841	0.576	0.890	0.932	0.833
PhiTa	1	0.693	0.883	0.591	0.918	0.904	0.901
	2	0.603	0.869	0.479	0.895	0.867	0.885
	3	0.389	0.865	0.193	0.897	0.889	0.879
	4	0.369	0.859	0.148	0.886	0.923	0.873
	5	0.334	0.854	0.252	0.879	0.796	0.873
	6	0.310	0.845	0.338	0.905	0.700	0.817
	7	0.325	0.847	0.301	0.901	0.746	0.830
	8	0.350	0.857	0.262	0.875	0.808	0.885
	9	0.337	0.857	0.303	0.890	0.762	0.870
	10	0.344	0.833	0.299	0.908	0.780	0.793
	11	0.355	0.829	0.256	0.869	0.824	0.836
	12	0.528	0.854	0.338	0.880	0.898	0.876
mT5-base	1	0.451	0.885	0.135	0.877	0.981	0.947
	2	0.435	0.874	0.110	0.890	0.975	0.910
	3	0.445	0.830	0.161	0.794	0.926	0.941
	4	0.467	0.798	0.269	0.787	0.858	0.882
	5	0.352	0.756	0.422	0.746	0.659	0.864
	6	0.346	0.775	0.636	0.766	0.489	0.876
	7	0.352	0.775	0.594	0.793	0.517	0.851
	8	0.366	0.784	0.557	0.809	0.560	0.842
	9	0.394	0.787	0.364	0.841	0.746	0.814
	10	0.387	0.788	0.314	0.794	0.783	0.876
	11	0.378	0.765	0.297	0.763	0.780	0.854
	12	0.466	0.779	0.198	0.834	0.926	0.799

Table 8: Per-layer Task B top- k ranking metrics for the three T5 encoders. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. This table is computed on the single-seed v2 split (no M -seed averaging, since the mseed sweep was run only for SIF-conditioned variants; see Table 16 for the multi-seed SIF+ABTT view). **Acc@1** is directory accuracy at rank 1, with an existing/new decomposition. Rows in bold mark the best layer per model, selected by ABTT Acc@1.

D Last-Token vs. Mean Pooling

Decoder-only embedding models often use last-token pooling. We compare last-token pooling with mean pooling under the same ABTT pipeline, train/test split, and ABTT choices. Table 9 reports the layer with the best Task B autonomous-routing accuracy, Task B directory accuracy at rank 1, and cosine gap at the same layer. Mean pooling performs better on assignment accuracy for every model in this experiment, including the decoder-only models.

E Per-Layer Results for Non-T5 Models

The main text focuses on LaTa, PhilTa, and mT5-base. This appendix reports the same per-layer view for LaBSE, Qwen3-0.6B, and KaLM-mini under the same comparison: baseline mean pooling versus ABTT with D tuned per layer on the train split.

Table 9: Mean vs. last-token pooling on the balanced split, restricted to ABTT post-processing. Best layer per (model, pooling) selected by Task A assignment accuracy. Task B metrics (Acc@1, cosine gap) reported at that same layer.

Model	Pool	Layer	Method	Assign Acc	Acc@1	Gap
LaTa	Mean	1	ABTT (D=10)	0.909	0.880	0.565
LaTa	LastTok	12	ABTT (D=10)	0.676	0.425	0.276
PhilTa	Mean	1	ABTT (D=10)	0.911	0.883	0.573
PhilTa	LastTok	9	ABTT (D=10)	0.604	0.385	0.200
LaBSE	Mean	11	ABTT (D=10)	0.908	0.885	0.610
LaBSE	LastTok	12	ABTT (D=10)	0.895	0.871	0.586
Qwen3-0.6B	Mean	5	ABTT (D=10)	0.915	0.894	0.545
Qwen3-0.6B	LastTok	27	ABTT (D=10)	0.907	0.885	0.562
KaLM-mini	Mean	1	ABTT (D=10)	0.921	0.902	0.568
KaLM-mini	LastTok	24	ABTT (D=10)	0.875	0.847	0.483
Qwen3-8B	Mean	1	ABTT (D=10)	0.924	0.897	0.543
Qwen3-8B	LastTok	35	ABTT (D=10)	0.910	0.892	0.642

Table 10: Per-layer Task A pairwise metrics for the non-T5 models (LaBSE, Qwen3-0.6B, KaLM-mini), under the same two-method comparison as the main paper. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. Both AUROC and cosine gap are computed over the test $n \times n$ cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the best layer per model, selected by ABTT AUROC.

Task A: Pairwise Duplicate Detection (appendix models)					
Model	Layer	AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
LaBSE	1	0.806	0.977	0.060	0.520
	2	0.833	0.979	0.037	0.531
	3	0.853	0.978	0.062	0.543
	4	0.862	0.977	0.073	0.525
	5	0.864	0.968	0.066	0.501
	6	0.894	0.976	0.078	0.532
	7	0.887	0.975	0.088	0.524
	8	0.879	0.975	0.080	0.522
	9	0.916	0.979	0.090	0.565
	10	0.934	0.982	0.103	0.586
	11	0.960	0.987	0.061	0.611
	12	0.956	0.981	0.183	0.592
Qwen3-0.6B	1	0.858	0.978	0.033	0.564
	2	0.863	0.973	0.024	0.553
	3	0.862	0.980	0.037	0.556
	4	0.860	0.986	0.036	0.547
	5	0.861	0.981	0.030	0.545
	6	0.867	0.983	0.035	0.545
	7	0.876	0.980	0.033	0.544
	8	0.882	0.972	0.037	0.532
	9	0.881	0.977	0.034	0.532
	10	0.887	0.976	0.031	0.529
	11	0.895	0.977	0.030	0.530
	12	0.901	0.973	0.027	0.530
	13	0.907	0.974	0.026	0.523
	14	0.905	0.974	0.024	0.519
	15	0.901	0.971	0.022	0.514
	16	0.910	0.975	0.026	0.509
	17	0.916	0.979	0.032	0.521
	18	0.933	0.983	0.032	0.537
	19	0.936	0.980	0.035	0.541
	20	0.943	0.982	0.041	0.559
	21	0.953	0.982	0.047	0.562
	22	0.963	0.983	0.044	0.583
	23	0.961	0.984	0.038	0.578
	24	0.963	0.984	0.032	0.576
	25	0.961	0.983	0.027	0.578
	26	0.966	0.985	0.024	0.568
	27	0.967	0.982	0.017	0.559
	28	0.958	0.975	0.098	0.556
KaLM-mini	1	0.887	0.981	0.063	0.568
	2	0.906	0.983	0.044	0.566
	3	0.881	0.981	0.062	0.551
	4	0.871	0.983	0.060	0.553
	5	0.861	0.983	0.062	0.541
	6	0.872	0.983	0.063	0.529
	7	0.888	0.979	0.072	0.524
	8	0.890	0.976	0.072	0.524
	9	0.892	0.974	0.066	0.507
	10	0.893	0.971	0.067	0.494
	11	0.894	0.970	0.061	0.475
	12	0.905	0.970	0.071	0.475
	13	0.900	0.968	0.066	0.467
	14	0.904	0.971	0.063	0.471
	15	0.907	0.973	0.067	0.468
	16	0.917	0.972	0.071	0.489

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Task A: Pairwise Duplicate Detection (appendix models)					
Model	Layer	AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
	17	0.926	0.975	0.075	0.501
	18	0.945	0.977	0.060	0.519
	19	0.956	0.979	0.061	0.531
	20	0.960	0.980	0.054	0.555
	21	0.962	0.981	0.043	0.555
	22	0.971	0.985	0.050	0.555
	23	0.972	0.986	0.056	0.552
	24	0.956	0.981	0.243	0.520

Table 11: Per-layer Task B autonomous routing accuracy for the non-T5 models. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. For every test file we compute $s_i = \max_{j \neq i} \cos(e_i, e_j)$ against the rest of the test set and compare to the learned threshold τ (fit on train via F_1 -optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if $s_i \geq \tau$, else new); only files predicted existing are then routed to their top-1 neighbour’s directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the best layer per model, selected by ABTT overall assignment accuracy.

Task B: Autonomous Routing (appendix models)							
Model	Layer	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
LaBSE	1	0.222	0.879	0.898	0.873	0.477	0.876
	2	0.381	0.875	0.811	0.827	0.543	0.857
	3	0.277	0.886	0.907	0.867	0.514	0.879
	4	0.372	0.897	0.885	0.836	0.565	0.874
	5	0.327	0.845	0.901	0.851	0.543	0.847
	6	0.548	0.860	0.882	0.851	0.674	0.857
	7	0.398	0.849	0.892	0.830	0.584	0.841
	8	0.385	0.869	0.870	0.743	0.568	0.822
	9	0.421	0.871	0.929	0.817	0.612	0.851
	10	0.606	0.892	0.892	0.805	0.713	0.859
	11	0.785	0.923	0.913	0.882	0.833	0.908
	12	0.729	0.914	0.938	0.864	0.808	0.895
Qwen3-0.6B	1	0.264	0.921	0.960	0.885	0.526	0.908
	2	0.437	0.869	0.814	0.916	0.579	0.887
	3	0.250	0.914	0.947	0.907	0.513	0.911
	4	0.411	0.936	0.848	0.842	0.576	0.901
	5	0.181	0.910	0.932	0.923	0.464	0.915
	6	0.422	0.935	0.830	0.796	0.576	0.882
	7	0.250	0.890	0.947	0.876	0.513	0.885
	8	0.409	0.931	0.895	0.799	0.592	0.881
	9	0.460	0.901	0.851	0.851	0.607	0.882
	10	0.533	0.892	0.836	0.864	0.647	0.881
	11	0.293	0.903	0.978	0.808	0.551	0.867
	12	0.407	0.912	0.938	0.817	0.607	0.876
	13	0.432	0.908	0.944	0.836	0.625	0.881
	14	0.456	0.907	0.926	0.811	0.633	0.871
	15	0.469	0.901	0.920	0.845	0.639	0.880
	16	0.639	0.897	0.830	0.808	0.711	0.864
	17	0.480	0.886	0.935	0.851	0.652	0.873
	18	0.553	0.882	0.944	0.839	0.700	0.866
	19	0.514	0.871	0.966	0.858	0.684	0.866
	20	0.654	0.865	0.898	0.892	0.746	0.875
	21	0.647	0.912	0.950	0.898	0.761	0.907
	22	0.731	0.888	0.938	0.916	0.809	0.899
	23	0.622	0.914	0.966	0.895	0.752	0.907
	24	0.748	0.901	0.935	0.935	0.818	0.914
	25	0.574	0.893	0.975	0.923	0.725	0.904
	26	0.613	0.899	0.975	0.895	0.749	0.897
	27	0.793	0.877	0.833	0.923	0.808	0.894

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Task B: Autonomous Routing (appendix models)							
Model	Layer	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
	28	0.796	0.877	0.867	0.913	0.823	0.890
KaLM-mini	1	0.391	0.921	0.907	0.920	0.585	0.921
	2	0.493	0.916	0.848	0.879	0.627	0.902
	3	0.443	0.923	0.870	0.907	0.604	0.917
	4	0.308	0.931	0.935	0.895	0.544	0.917
	5	0.338	0.905	0.882	0.923	0.543	0.911
	6	0.256	0.895	0.910	0.920	0.502	0.904
	7	0.475	0.864	0.854	0.944	0.618	0.894
	8	0.467	0.862	0.870	0.913	0.619	0.881
	9	0.469	0.873	0.885	0.870	0.626	0.872
	10	0.449	0.875	0.885	0.870	0.613	0.873
	11	0.458	0.862	0.885	0.864	0.619	0.862
	12	0.507	0.862	0.895	0.904	0.653	0.878
	13	0.540	0.860	0.895	0.870	0.674	0.864
	14	0.593	0.836	0.845	0.885	0.688	0.854
	15	0.566	0.864	0.898	0.827	0.691	0.850
	16	0.550	0.849	0.935	0.830	0.695	0.841
	17	0.675	0.864	0.920	0.889	0.767	0.873
	18	0.697	0.875	0.960	0.898	0.796	0.883
	19	0.721	0.873	0.966	0.892	0.814	0.880
	20	0.705	0.791	0.966	0.913	0.803	0.837
	21	0.727	0.811	0.966	0.926	0.817	0.854
	22	0.850	0.879	0.916	0.923	0.875	0.895
	23	0.849	0.884	0.920	0.898	0.875	0.889
	24	0.787	0.892	0.944	0.867	0.846	0.882

Table 12: Per-layer Task B top- k ranking metrics for the non-T5 models. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. This table is computed on the single-seed v2 split (no M -seed averaging, since the mseed sweep was run only for SIF-conditioned variants; see Table 16 for the multi-seed SIF+ABTT view). **Acc@1** is directory accuracy at rank 1, with an existing/new decomposition. Rows in bold mark the best layer per model, selected by ABTT Acc@1.

Task B: Top-K Ranking (appendix models, single-seed)							
Model	Layer	Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
LaBSE	1	0.465	0.851	0.222	0.879	0.898	0.873
	2	0.494	0.830	0.381	0.875	0.811	0.827
	3	0.500	0.848	0.277	0.886	0.907	0.867
	4	0.542	0.841	0.372	0.897	0.885	0.836
	5	0.521	0.811	0.327	0.845	0.901	0.851
	6	0.640	0.831	0.548	0.860	0.882	0.851
	7	0.561	0.810	0.398	0.849	0.892	0.830
	8	0.544	0.779	0.385	0.869	0.870	0.743
	9	0.608	0.823	0.421	0.871	0.929	0.817
	10	0.693	0.832	0.606	0.892	0.892	0.805
	11	0.815	0.885	0.785	0.923	0.913	0.882
	12	0.791	0.868	0.729	0.914	0.938	0.864
Qwen3-0.6B	1	0.521	0.875	0.264	0.921	0.960	0.885
	2	0.558	0.871	0.437	0.869	0.814	0.916
	3	0.508	0.890	0.250	0.914	0.947	0.907
	4	0.559	0.874	0.411	0.936	0.848	0.842
	5	0.450	0.894	0.181	0.910	0.932	0.923
	6	0.551	0.858	0.422	0.935	0.830	0.796
	7	0.507	0.859	0.250	0.890	0.947	0.876
	8	0.580	0.852	0.409	0.931	0.895	0.799
	9	0.587	0.852	0.460	0.901	0.851	0.851
	10	0.622	0.848	0.533	0.892	0.836	0.864
	11	0.548	0.830	0.293	0.903	0.978	0.808
	12	0.603	0.848	0.407	0.912	0.938	0.817
	13	0.618	0.851	0.432	0.908	0.944	0.836

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Task B: Top-K Ranking (appendix models, single-seed)							
Model	Layer	Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
	14	0.621	0.840	0.456	0.907	0.926	0.811
	15	0.627	0.848	0.469	0.901	0.920	0.845
	16	0.682	0.830	0.639	0.897	0.830	0.808
	17	0.641	0.840	0.480	0.886	0.935	0.851
	18	0.691	0.837	0.553	0.882	0.944	0.839
	19	0.676	0.837	0.514	0.871	0.966	0.858
	20	0.733	0.852	0.654	0.865	0.898	0.892
	21	0.752	0.881	0.647	0.912	0.950	0.898
	22	0.795	0.876	0.731	0.888	0.938	0.916
	23	0.742	0.886	0.622	0.914	0.966	0.895
	24	0.808	0.896	0.748	0.901	0.935	0.935
	25	0.718	0.886	0.574	0.893	0.975	0.923
	26	0.740	0.881	0.613	0.899	0.975	0.895
	27	0.787	0.876	0.793	0.877	0.833	0.923
	28	0.803	0.872	0.796	0.877	0.867	0.913
	KaLM-mini 1	0.577	0.902	0.391	0.921	0.907	0.920
	2	0.607	0.879	0.493	0.916	0.848	0.879
	3	0.587	0.894	0.443	0.923	0.870	0.907
	4	0.538	0.887	0.308	0.931	0.935	0.895
	5	0.524	0.882	0.338	0.905	0.882	0.923
	6	0.490	0.876	0.256	0.895	0.910	0.920
	7	0.592	0.873	0.475	0.864	0.854	0.944
	8	0.597	0.860	0.467	0.862	0.870	0.913
	9	0.601	0.843	0.469	0.873	0.885	0.870
	10	0.584	0.844	0.449	0.875	0.885	0.870
	11	0.586	0.836	0.458	0.862	0.885	0.864
	12	0.622	0.850	0.507	0.862	0.895	0.904
	13	0.633	0.840	0.540	0.860	0.895	0.870
	14	0.636	0.829	0.593	0.836	0.845	0.885
	15	0.650	0.817	0.566	0.864	0.898	0.827
	16	0.676	0.819	0.550	0.849	0.935	0.830
	17	0.747	0.852	0.675	0.864	0.920	0.889
	18	0.777	0.865	0.697	0.875	0.960	0.898
	19	0.801	0.861	0.721	0.873	0.966	0.892
	20	0.789	0.817	0.705	0.791	0.966	0.913
	21	0.802	0.831	0.727	0.811	0.966	0.926
	22	0.857	0.875	0.850	0.879	0.916	0.923
	23	0.859	0.872	0.849	0.884	0.920	0.898
	24	0.833	0.858	0.787	0.892	0.944	0.867

F SIF and Whitening Post-Processing Sweeps

The main text uses ABTT-only as the paper-facing repair. This appendix reports SIF-only, SIF plus fixed- D ABTT, and SIF plus tuned- D ABTT, and gives the full account of the PCA whitening variant that Section 4.3 excludes. Table 13 first gives the five-seed cumulative top- K result discussed in the main text, with SIF plus ABTT at each model’s train-selected layer. The per-layer tables then keep one primary metric per task so the four-method grid stays readable.

Why whitening is excluded. PCA whitening (Su et al., 2021) leaves the different-directory cosine mean at or below 0.015, so the learned threshold degenerates: across all 100 whitening cells every file is routed to an existing directory, existing accuracy is 1.000, new accuracy never exceeds one file in 323, and assignment accuracy sits at the 62.4 percent class prior. That prior beats baseline in 60 of 100 cells while directory accuracy at rank 1 never exceeds 0.415, so the apparent gain is a thresholding artifact. SIF and whitening fall short of ABTT for different reasons. SIF acts before pooling, so it can only attenuate the frequent tokens that feed the dominant direction, and whatever survives that reweighting is still pooled into the passage vector; whitening acts after pooling but rescales all directions instead of deleting the top few, which flattens the same-directory margin along with the common component and so degenerates the routing threshold.

Model	Top-1	Top-2	Top-3	Top-4	Top-5
LaTa	88.9 $\pm$ 0.5	96.1 $\pm$ 0.5	97.4 $\pm$ 0.6	97.9 $\pm$ 0.3	98.2 $\pm$ 0.2
PhilTa	89.8 $\pm$ 0.8	95.8 $\pm$ 0.6	97.4 $\pm$ 0.3	97.7 $\pm$ 0.3	97.8 $\pm$ 0.4
mT5-base	89.3 $\pm$ 0.5	95.1 $\pm$ 0.2	96.7 $\pm$ 0.7	97.1 $\pm$ 0.4	97.2 $\pm$ 0.4
LaBSE	90.6 $\pm$ 1.0	95.5 $\pm$ 0.5	96.4 $\pm$ 0.5	97.2 $\pm$ 0.6	97.8 $\pm$ 0.5
Qwen3-0.6B	89.3 $\pm$ 0.7	95.7 $\pm$ 0.4	97.1 $\pm$ 0.3	97.5 $\pm$ 0.4	97.6 $\pm$ 0.4
KaLM-mini	89.3 $\pm$ 0.5	95.1 $\pm$ 0.4	96.6 $\pm$ 0.7	97.2 $\pm$ 0.7	97.6 $\pm$ 0.5

Table 13: Cumulative top- K accuracy for Task B under SIF+ABTT, as mean $\pm$ standard deviation over five query/reference reseeds at a fixed v2 split. For each model the reported layer is chosen on the train split: the layer with the highest training-set directory accuracy at rank 1 for SIF+ABTT in the single-seed run, the same layer as the SIF+ABTT subscript in Table 3. D is tuned per layer on the train split. This is not always the layer with the highest five-seed mean. Each row is the bold row of Table 16. Each cell is the percentage of test queries whose labelled directory appears within the top K options. Per-layer top- k rankings: Appendix Tables 8, 12, and 16.

Table 14: Per-layer Task A AUROC across the SIF-conditioned post-processing suite for all six models. `sif_only` replaces mean pooling with SIF-weighted pooling; `sif_abtt_fixed` adds ABTT with fixed $D=10$; `sif_abtt_optimal` tunes D per layer on the train split. Gap is omitted to keep the table narrow; the pure-ABTT comparison (not SIF-conditioned) is in Tables 6 and 10. Rows in bold mark the best layer per model, selected by SIF+ABTT (D-tuned) AUROC.

Model	Layer	Task A: SIF-suite AUROC (appendix)			
		AUROC (base)	AUROC (SIF)	AUROC (SIF+ABTT ₁₀)	AUROC (SIF+ABTT)
LaTa	1	0.934	0.948	0.972	0.976
	2	0.563	0.938	0.964	0.971
	3	0.513	0.921	0.967	0.966
	4	0.500	0.890	0.970	0.968
	5	0.496	0.878	0.970	0.970
	6	0.496	0.865	0.965	0.965
	7	0.498	0.858	0.966	0.966
	8	0.502	0.842	0.964	0.964
	9	0.503	0.854	0.964	0.964
	10	0.505	0.859	0.966	0.966
	11	0.508	0.867	0.965	0.965
	12	0.938	0.948	0.963	0.963
PhilTa	1	0.939	0.955	0.975	0.975
	2	0.911	0.941	0.973	0.973
	3	0.601	0.928	0.968	0.968
	4	0.551	0.916	0.972	0.972
	5	0.542	0.904	0.972	0.972
	6	0.541	0.884	0.981	0.981
	7	0.542	0.891	0.981	0.981
	8	0.542	0.895	0.982	0.982
	9	0.539	0.881	0.977	0.977
	10	0.538	0.874	0.977	0.977
	11	0.539	0.872	0.979	0.979
	12	0.911	0.929	0.969	0.969
mT5-base	1	0.822	0.845	0.974	0.974
	2	0.822	0.835	0.966	0.966
	3	0.804	0.808	0.958	0.958
	4	0.799	0.809	0.966	0.966
	5	0.654	0.672	0.974	0.974
	6	0.657	0.678	0.971	0.971
	7	0.656	0.680	0.967	0.967
	8	0.656	0.681	0.968	0.968
	9	0.660	0.685	0.967	0.967
	10	0.659	0.684	0.966	0.966
	11	0.659	0.689	0.966	0.966
	12	0.838	0.833	0.970	0.970
LaBSE	1	0.806	0.818	0.974	0.974
	2	0.833	0.840	0.975	0.975
	3	0.853	0.862	0.968	0.968
	4	0.862	0.869	0.966	0.966
	5	0.864	0.869	0.964	0.964
	6	0.894	0.895	0.965	0.965
	7	0.887	0.892	0.964	0.964
	8	0.879	0.883	0.968	0.968
	9	0.916	0.915	0.973	0.973
	10	0.934	0.934	0.976	0.976
	11	0.960	0.959	0.983	0.983
	12	0.956	0.956	0.980	0.984
Qwen3-0.6B	1	0.858	0.852	0.982	0.981
	2	0.863	0.853	0.972	0.972
	3	0.862	0.858	0.977	0.975
	4	0.860	0.860	0.976	0.977
	5	0.861	0.858	0.971	0.971
	6	0.867	0.861	0.974	0.974
	7	0.876	0.866	0.971	0.971
	8	0.882	0.868	0.969	0.969

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Task A: SIF-suite AUROC (appendix)					
Model	Layer	AUROC (base)	AUROC (SIF)	AUROC (SIF+ABTT ₁₀)	AUROC (SIF+ABTT)
	9	0.881	0.866	0.968	0.968
	10	0.887	0.870	0.966	0.966
	11	0.895	0.875	0.964	0.964
	12	0.901	0.881	0.966	0.966
	13	0.907	0.887	0.967	0.967
	14	0.905	0.885	0.966	0.966
	15	0.901	0.883	0.968	0.968
	16	0.910	0.894	0.965	0.965
	17	0.916	0.899	0.967	0.974
	18	0.933	0.914	0.966	0.973
	19	0.936	0.915	0.966	0.976
	20	0.943	0.918	0.971	0.977
	21	0.953	0.931	0.971	0.971
	22	0.963	0.942	0.973	0.973
	23	0.961	0.941	0.973	0.973
	24	0.963	0.943	0.976	0.974
	25	0.961	0.938	0.978	0.975
	26	0.966	0.942	0.980	0.980
	27	0.967	0.936	0.978	0.978
	28	0.958	0.925	0.968	0.968
	KaLM-mini 1	0.887	0.880	0.982	0.980
	2	0.906	0.899	0.981	0.981
	3	0.881	0.882	0.977	0.977
	4	0.871	0.873	0.974	0.974
	5	0.861	0.862	0.971	0.971
	6	0.872	0.867	0.967	0.967
	7	0.888	0.879	0.972	0.972
	8	0.890	0.880	0.973	0.975
	9	0.892	0.882	0.968	0.968
	10	0.893	0.882	0.964	0.964
	11	0.894	0.885	0.969	0.969
	12	0.905	0.896	0.966	0.966
	13	0.900	0.894	0.964	0.964
	14	0.904	0.898	0.970	0.970
	15	0.907	0.901	0.970	0.970
	16	0.917	0.913	0.967	0.967
	17	0.926	0.923	0.968	0.968
	18	0.945	0.939	0.977	0.977
	19	0.956	0.950	0.978	0.978
	20	0.960	0.956	0.978	0.978
	21	0.962	0.958	0.980	0.983
	22	0.971	0.967	0.983	0.984
	23	0.972	0.964	0.986	0.985
	24	0.956	0.956	0.976	0.976

Table 15: Per-layer Task B overall assignment accuracy across the SIF-conditioned suite for all six models. Restricted to overall routing accuracy (existing/new decomposition omitted) to keep the table compact; see Tables 7 and 11 for the pure-ABTT pairwise comparison with existing/new broken out. Rows in bold mark the best layer per model, selected by SIF+ABTT (D-tuned) overall assignment accuracy.

Task B: SIF-suite routing (appendix)					
Model	Layer	Overall (base)	Overall (SIF)	Overall (SIF+ABTT ₁₀)	Overall (SIF+ABTT)
LaTa	1	0.738	0.832	0.920	0.899
	2	0.439	0.791	0.904	0.900
	3	0.477	0.720	0.910	0.907
	4	0.513	0.668	0.900	0.902
	5	0.502	0.648	0.906	0.906
	6	0.503	0.668	0.895	0.895
	7	0.501	0.642	0.903	0.903
	8	0.502	0.614	0.892	0.892

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Task B: SIF-suite routing (appendix)					
Model	Layer	Overall (base)	Overall (SIF)	Overall (SIF+ABTT ₁₀)	Overall (SIF+ABTT)
	9	0.501	0.618	0.889	0.889
	10	0.498	0.655	0.890	0.890
	11	0.490	0.621	0.880	0.880
	12	0.710	0.804	0.902	0.902
PhilTa	1	0.709	0.790	0.914	0.914
	2	0.625	0.738	0.892	0.892
	3	0.455	0.713	0.892	0.892
	4	0.439	0.663	0.886	0.886
	5	0.457	0.626	0.899	0.899
	6	0.474	0.612	0.906	0.906
	7	0.469	0.625	0.906	0.906
	8	0.467	0.614	0.886	0.886
	9	0.476	0.590	0.895	0.895
	10	0.480	0.580	0.888	0.888
	11	0.470	0.565	0.886	0.886
	12	0.549	0.667	0.899	0.899
mT5-base	1	0.453	0.545	0.907	0.907
	2	0.436	0.537	0.903	0.903
	3	0.449	0.568	0.885	0.885
	4	0.491	0.488	0.857	0.857
	5	0.512	0.483	0.826	0.826
	6	0.580	0.558	0.817	0.817
	7	0.565	0.554	0.832	0.832
	8	0.558	0.548	0.834	0.834
	9	0.508	0.495	0.845	0.845
	10	0.491	0.495	0.832	0.832
	11	0.479	0.484	0.816	0.816
	12	0.472	0.541	0.858	0.858
LaBSE	1	0.477	0.521	0.886	0.886
	2	0.543	0.508	0.875	0.875
	3	0.514	0.577	0.860	0.860
	4	0.565	0.587	0.869	0.869
	5	0.543	0.598	0.857	0.857
	6	0.674	0.648	0.872	0.872
	7	0.584	0.634	0.858	0.858
	8	0.568	0.571	0.854	0.854
	9	0.612	0.613	0.859	0.859
	10	0.713	0.698	0.864	0.864
	11	0.833	0.807	0.896	0.896
	12	0.808	0.834	0.890	0.880
Qwen3-0.6B	1	0.526	0.566	0.914	0.909
	2	0.579	0.524	0.894	0.894
	3	0.513	0.537	0.904	0.901
	4	0.576	0.531	0.899	0.904
	5	0.464	0.537	0.908	0.908
	6	0.576	0.526	0.906	0.906
	7	0.513	0.573	0.911	0.911
	8	0.592	0.627	0.911	0.911
	9	0.607	0.576	0.910	0.910
	10	0.647	0.599	0.900	0.900
	11	0.551	0.606	0.893	0.893
	12	0.607	0.663	0.900	0.900
	13	0.625	0.675	0.897	0.897
	14	0.633	0.571	0.893	0.893
	15	0.639	0.600	0.893	0.893
	16	0.711	0.679	0.880	0.880
	17	0.652	0.686	0.900	0.876
	18	0.700	0.731	0.899	0.882
	19	0.684	0.700	0.886	0.850
	20	0.746	0.676	0.882	0.866
	21	0.761	0.739	0.892	0.892
	22	0.809	0.719	0.893	0.893

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Task B: SIF-suite routing (appendix)					
Model	Layer	Overall (base)	Overall (SIF)	Overall (SIF+ABTT ₁₀)	Overall (SIF+ABTT)
KaLM-mini	23	0.752	0.759	0.893	0.893
	24	0.818	0.706	0.900	0.899
	25	0.725	0.748	0.897	0.893
	26	0.749	0.772	0.903	0.903
	27	0.808	0.760	0.899	0.899
	28	0.823	0.774	0.890	0.890
	1	0.585	0.604	0.906	0.908
	2	0.627	0.594	0.908	0.908
	3	0.604	0.576	0.907	0.907
	4	0.544	0.577	0.901	0.901
	5	0.543	0.533	0.897	0.897
	6	0.502	0.558	0.900	0.900
	7	0.618	0.593	0.901	0.901
	8	0.619	0.607	0.892	0.878
	9	0.626	0.663	0.876	0.876
	10	0.613	0.603	0.866	0.866
	11	0.619	0.627	0.874	0.874
	12	0.653	0.642	0.874	0.874
	13	0.674	0.666	0.864	0.864
	14	0.688	0.648	0.861	0.861
	15	0.691	0.698	0.866	0.866
	16	0.695	0.738	0.859	0.859
	17	0.767	0.763	0.882	0.882
	18	0.796	0.796	0.879	0.879
	19	0.814	0.804	0.878	0.878
	20	0.803	0.821	0.876	0.876
	21	0.817	0.832	0.874	0.876
	22	0.875	0.857	0.876	0.876
	23	0.875	0.862	0.880	0.881
	24	0.846	0.847	0.869	0.869

Table 16: Per-layer Task B top- k ranking accuracy across all six models, averaged over $M = 5$ query/reference reseeds of the v2 train/test split (mean $\pm$ std). `sif_abtt_optimal` applies SIF weighting plus ABTT with D tuned per layer on the train split; the mseed sweep was not run for the ABTT-only variant of Tables 2 and 3, so we include this table to give a variance-aware view of the SIF-conditioned best variant. Overall assignment accuracy coincides with Acc@1 in the mseed pipeline, so we report the existing/new decomposition instead. Rows in bold mark the layer reported per model, the row printed in Table 13. For each model the reported layer is chosen on the train split: the layer with the highest training-set directory accuracy at rank 1 for SIF+ABTT in the single-seed run, the same layer as the SIF+ABTT subscript in Table 3. D is tuned per layer on the train split. This is not always the layer with the highest five-seed mean.

Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT)							
Model	Layer	Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
LaTa	1	0.731 $\pm$ 0.010	0.889 $\pm$ 0.010	0.563 $\pm$ 0.007	0.844 $\pm$ 0.011	0.949 $\pm$ 0.011	0.948 $\pm$ 0.010
	2	0.429 $\pm$ 0.011	0.888 $\pm$ 0.007	0.049 $\pm$ 0.006	0.841 $\pm$ 0.007	0.920 $\pm$ 0.008	0.948 $\pm$ 0.007
	3	0.393 $\pm$ 0.011	0.891 $\pm$ 0.010	0.047 $\pm$ 0.006	0.847 $\pm$ 0.010	0.840 $\pm$ 0.020	0.947 $\pm$ 0.012
	4	0.331 $\pm$ 0.019	0.889 $\pm$ 0.009	0.048 $\pm$ 0.009	0.860 $\pm$ 0.009	0.696 $\pm$ 0.030	0.925 $\pm$ 0.013
	5	0.326 $\pm$ 0.021	0.891 $\pm$ 0.009	0.038 $\pm$ 0.005	0.858 $\pm$ 0.008	0.698 $\pm$ 0.028	0.934 $\pm$ 0.014
	6	0.322 $\pm$ 0.018	0.891 $\pm$ 0.009	0.038 $\pm$ 0.007	0.863 $\pm$ 0.010	0.688 $\pm$ 0.026	0.927 $\pm$ 0.013
	7	0.328 $\pm$ 0.020	0.896 $\pm$ 0.008	0.045 $\pm$ 0.007	0.848 $\pm$ 0.009	0.693 $\pm$ 0.028	0.957 $\pm$ 0.010
	8	0.325 $\pm$ 0.022	0.884 $\pm$ 0.008	0.046 $\pm$ 0.010	0.834 $\pm$ 0.008	0.685 $\pm$ 0.032	0.949 $\pm$ 0.011
	9	0.332 $\pm$ 0.019	0.886 $\pm$ 0.011	0.045 $\pm$ 0.010	0.848 $\pm$ 0.009	0.701 $\pm$ 0.024	0.935 $\pm$ 0.015
	10	0.346 $\pm$ 0.017	0.889 $\pm$ 0.004	0.045 $\pm$ 0.010	0.846 $\pm$ 0.008	0.735 $\pm$ 0.021	0.944 $\pm$ 0.012
	11	0.361 $\pm$ 0.018	0.882 $\pm$ 0.009	0.045 $\pm$ 0.009	0.862 $\pm$ 0.010	0.767 $\pm$ 0.029	0.907 $\pm$ 0.014
	12	0.707 $\pm$ 0.014	0.889 $\pm$ 0.005	0.511 $\pm$ 0.016	0.834 $\pm$ 0.005	0.961 $\pm$ 0.008	0.961 $\pm$ 0.010
PhilTa	1	0.705 $\pm$ 0.015	0.898 $\pm$ 0.008	0.531 $\pm$ 0.016	0.858 $\pm$ 0.010	0.931 $\pm$ 0.012	0.951 $\pm$ 0.011
	2	0.626 $\pm$ 0.019	0.888 $\pm$ 0.010	0.410 $\pm$ 0.017	0.866 $\pm$ 0.010	0.906 $\pm$ 0.018	0.917 $\pm$ 0.017
	3	0.440 $\pm$ 0.013	0.893 $\pm$ 0.009	0.071 $\pm$ 0.014	0.858 $\pm$ 0.010	0.917 $\pm$ 0.013	0.939 $\pm$ 0.011

(continued on next page)

Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT)							
Model	Layer	Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
	4	0.421 $\pm$ 0.016	0.891 $\pm$ 0.009	0.027 $\pm$ 0.007	0.858 $\pm$ 0.008	0.928 $\pm$ 0.015	0.935 $\pm$ 0.013
	5	0.375 $\pm$ 0.016	0.895 $\pm$ 0.010	0.042 $\pm$ 0.013	0.852 $\pm$ 0.010	0.804 $\pm$ 0.018	0.950 $\pm$ 0.014
	6	0.337 $\pm$ 0.015	0.897 $\pm$ 0.009	0.060 $\pm$ 0.012	0.861 $\pm$ 0.010	0.696 $\pm$ 0.026	0.944 $\pm$ 0.013
	7	0.361 $\pm$ 0.017	0.900 $\pm$ 0.009	0.056 $\pm$ 0.012	0.860 $\pm$ 0.008	0.755 $\pm$ 0.020	0.952 $\pm$ 0.013
	8	0.389 $\pm$ 0.016	0.891 $\pm$ 0.010	0.058 $\pm$ 0.014	0.859 $\pm$ 0.009	0.818 $\pm$ 0.018	0.932 $\pm$ 0.015
	9	0.371 $\pm$ 0.016	0.893 $\pm$ 0.008	0.066 $\pm$ 0.014	0.861 $\pm$ 0.009	0.766 $\pm$ 0.027	0.934 $\pm$ 0.016
	10	0.377 $\pm$ 0.014	0.892 $\pm$ 0.009	0.066 $\pm$ 0.014	0.855 $\pm$ 0.005	0.778 $\pm$ 0.023	0.940 $\pm$ 0.017
	11	0.394 $\pm$ 0.014	0.888 $\pm$ 0.008	0.059 $\pm$ 0.010	0.851 $\pm$ 0.010	0.827 $\pm$ 0.021	0.935 $\pm$ 0.013
	12	0.566 $\pm$ 0.013	0.897 $\pm$ 0.010	0.282 $\pm$ 0.014	0.863 $\pm$ 0.010	0.934 $\pm$ 0.007	0.941 $\pm$ 0.018
	1	0.487 $\pm$ 0.012	0.899 $\pm$ 0.005	0.100 $\pm$ 0.009	0.856 $\pm$ 0.007	0.987 $\pm$ 0.007	0.954 $\pm$ 0.011
	2	0.476 $\pm$ 0.012	0.893 $\pm$ 0.005	0.083 $\pm$ 0.008	0.851 $\pm$ 0.012	0.984 $\pm$ 0.008	0.947 $\pm$ 0.013
	3	0.483 $\pm$ 0.014	0.883 $\pm$ 0.011	0.124 $\pm$ 0.009	0.817 $\pm$ 0.014	0.947 $\pm$ 0.008	0.970 $\pm$ 0.007
mT5-base	4	0.500 $\pm$ 0.015	0.852 $\pm$ 0.007	0.206 $\pm$ 0.006	0.772 $\pm$ 0.010	0.879 $\pm$ 0.020	0.956 $\pm$ 0.010
	5	0.371 $\pm$ 0.014	0.825 $\pm$ 0.008	0.155 $\pm$ 0.006	0.724 $\pm$ 0.007	0.651 $\pm$ 0.023	0.956 $\pm$ 0.006
	6	0.361 $\pm$ 0.012	0.810 $\pm$ 0.005	0.263 $\pm$ 0.021	0.765 $\pm$ 0.012	0.488 $\pm$ 0.018	0.868 $\pm$ 0.016
	7	0.370 $\pm$ 0.010	0.827 $\pm$ 0.008	0.255 $\pm$ 0.014	0.760 $\pm$ 0.011	0.519 $\pm$ 0.017	0.914 $\pm$ 0.013
	8	0.382 $\pm$ 0.008	0.824 $\pm$ 0.012	0.244 $\pm$ 0.007	0.773 $\pm$ 0.012	0.561 $\pm$ 0.017	0.891 $\pm$ 0.016
	9	0.428 $\pm$ 0.014	0.838 $\pm$ 0.010	0.171 $\pm$ 0.008	0.783 $\pm$ 0.013	0.759 $\pm$ 0.024	0.909 $\pm$ 0.006
	10	0.421 $\pm$ 0.010	0.821 $\pm$ 0.017	0.140 $\pm$ 0.011	0.767 $\pm$ 0.015	0.783 $\pm$ 0.021	0.892 $\pm$ 0.019
	11	0.420 $\pm$ 0.015	0.803 $\pm$ 0.012	0.131 $\pm$ 0.012	0.721 $\pm$ 0.013	0.792 $\pm$ 0.016	0.910 $\pm$ 0.015
	12	0.507 $\pm$ 0.009	0.841 $\pm$ 0.009	0.169 $\pm$ 0.007	0.772 $\pm$ 0.016	0.942 $\pm$ 0.011	0.931 $\pm$ 0.011
	1	0.502 $\pm$ 0.012	0.858 $\pm$ 0.007	0.170 $\pm$ 0.003	0.838 $\pm$ 0.007	0.929 $\pm$ 0.016	0.885 $\pm$ 0.009
	2	0.523 $\pm$ 0.011	0.860 $\pm$ 0.004	0.274 $\pm$ 0.005	0.817 $\pm$ 0.003	0.844 $\pm$ 0.022	0.915 $\pm$ 0.008
	3	0.535 $\pm$ 0.010	0.847 $\pm$ 0.006	0.218 $\pm$ 0.009	0.831 $\pm$ 0.009	0.945 $\pm$ 0.008	0.867 $\pm$ 0.007
LaBSE	4	0.569 $\pm$ 0.008	0.850 $\pm$ 0.006	0.299 $\pm$ 0.004	0.837 $\pm$ 0.006	0.917 $\pm$ 0.014	0.866 $\pm$ 0.011
	5	0.551 $\pm$ 0.012	0.845 $\pm$ 0.005	0.255 $\pm$ 0.012	0.794 $\pm$ 0.005	0.933 $\pm$ 0.013	0.911 $\pm$ 0.010
	6	0.661 $\pm$ 0.011	0.859 $\pm$ 0.004	0.461 $\pm$ 0.019	0.817 $\pm$ 0.008	0.919 $\pm$ 0.018	0.914 $\pm$ 0.015
	7	0.596 $\pm$ 0.010	0.849 $\pm$ 0.004	0.333 $\pm$ 0.016	0.808 $\pm$ 0.006	0.935 $\pm$ 0.016	0.902 $\pm$ 0.010
	8	0.572 $\pm$ 0.012	0.847 $\pm$ 0.006	0.314 $\pm$ 0.019	0.802 $\pm$ 0.009	0.904 $\pm$ 0.016	0.905 $\pm$ 0.008
	9	0.631 $\pm$ 0.008	0.854 $\pm$ 0.007	0.381 $\pm$ 0.015	0.828 $\pm$ 0.002	0.954 $\pm$ 0.008	0.887 $\pm$ 0.012
	10	0.712 $\pm$ 0.016	0.858 $\pm$ 0.005	0.551 $\pm$ 0.019	0.836 $\pm$ 0.005	0.920 $\pm$ 0.015	0.887 $\pm$ 0.010
	11	0.820 $\pm$ 0.010	0.906 $\pm$ 0.010	0.731 $\pm$ 0.012	0.875 $\pm$ 0.013	0.935 $\pm$ 0.012	0.944 $\pm$ 0.017
	12	0.793 $\pm$ 0.015	0.884 $\pm$ 0.006	0.670 $\pm$ 0.019	0.850 $\pm$ 0.012	0.952 $\pm$ 0.013	0.927 $\pm$ 0.015
	1	0.550 $\pm$ 0.008	0.894 $\pm$ 0.005	0.223 $\pm$ 0.006	0.840 $\pm$ 0.010	0.972 $\pm$ 0.003	0.965 $\pm$ 0.011
	2	0.572 $\pm$ 0.015	0.882 $\pm$ 0.004	0.363 $\pm$ 0.014	0.828 $\pm$ 0.005	0.841 $\pm$ 0.032	0.952 $\pm$ 0.007
	3	0.541 $\pm$ 0.007	0.889 $\pm$ 0.006	0.216 $\pm$ 0.007	0.827 $\pm$ 0.008	0.962 $\pm$ 0.007	0.968 $\pm$ 0.011
Qwen3-0.6B	4	0.580 $\pm$ 0.012	0.891 $\pm$ 0.004	0.353 $\pm$ 0.008	0.845 $\pm$ 0.007	0.874 $\pm$ 0.028	0.950 $\pm$ 0.006
	5	0.487 $\pm$ 0.017	0.893 $\pm$ 0.007	0.133 $\pm$ 0.012	0.839 $\pm$ 0.011	0.944 $\pm$ 0.014	0.964 $\pm$ 0.009
	6	0.568 $\pm$ 0.012	0.897 $\pm$ 0.005	0.350 $\pm$ 0.010	0.857 $\pm$ 0.008	0.850 $\pm$ 0.023	0.948 $\pm$ 0.012
	7	0.541 $\pm$ 0.007	0.902 $\pm$ 0.006	0.211 $\pm$ 0.009	0.862 $\pm$ 0.008	0.967 $\pm$ 0.012	0.953 $\pm$ 0.012
	8	0.609 $\pm$ 0.009	0.903 $\pm$ 0.006	0.362 $\pm$ 0.013	0.861 $\pm$ 0.010	0.929 $\pm$ 0.017	0.956 $\pm$ 0.005
	9	0.612 $\pm$ 0.013	0.900 $\pm$ 0.004	0.393 $\pm$ 0.016	0.868 $\pm$ 0.008	0.894 $\pm$ 0.021	0.943 $\pm$ 0.006
	10	0.638 $\pm$ 0.013	0.890 $\pm$ 0.009	0.452 $\pm$ 0.009	0.847 $\pm$ 0.014	0.878 $\pm$ 0.022	0.944 $\pm$ 0.010
	11	0.578 $\pm$ 0.006	0.885 $\pm$ 0.005	0.257 $\pm$ 0.010	0.838 $\pm$ 0.014	0.991 $\pm$ 0.005	0.947 $\pm$ 0.012
	12	0.627 $\pm$ 0.009	0.892 $\pm$ 0.005	0.366 $\pm$ 0.016	0.844 $\pm$ 0.010	0.964 $\pm$ 0.013	0.955 $\pm$ 0.007
	13	0.639 $\pm$ 0.010	0.887 $\pm$ 0.006	0.383 $\pm$ 0.014	0.849 $\pm$ 0.011	0.970 $\pm$ 0.015	0.937 $\pm$ 0.011
	14	0.646 $\pm$ 0.009	0.885 $\pm$ 0.006	0.402 $\pm$ 0.017	0.852 $\pm$ 0.009	0.960 $\pm$ 0.016	0.927 $\pm$ 0.007
	15	0.648 $\pm$ 0.012	0.885 $\pm$ 0.007	0.410 $\pm$ 0.017	0.852 $\pm$ 0.010	0.954 $\pm$ 0.017	0.929 $\pm$ 0.018
	16	0.691 $\pm$ 0.016	0.873 $\pm$ 0.005	0.546 $\pm$ 0.017	0.837 $\pm$ 0.009	0.878 $\pm$ 0.017	0.920 $\pm$ 0.016
	17	0.658 $\pm$ 0.010	0.862 $\pm$ 0.006	0.420 $\pm$ 0.018	0.807 $\pm$ 0.012	0.966 $\pm$ 0.013	0.933 $\pm$ 0.013
	18	0.709 $\pm$ 0.012	0.868 $\pm$ 0.002	0.503 $\pm$ 0.018	0.833 $\pm$ 0.013	0.975 $\pm$ 0.006	0.914 $\pm$ 0.017
	19	0.691 $\pm$ 0.010	0.874 $\pm$ 0.002	0.462 $\pm$ 0.018	0.860 $\pm$ 0.007	0.986 $\pm$ 0.006	0.893 $\pm$ 0.012
	20	0.743 $\pm$ 0.020	0.860 $\pm$ 0.004	0.588 $\pm$ 0.023	0.797 $\pm$ 0.011	0.943 $\pm$ 0.018	0.941 $\pm$ 0.010
	21	0.758 $\pm$ 0.017	0.879 $\pm$ 0.006	0.585 $\pm$ 0.021	0.836 $\pm$ 0.008	0.981 $\pm$ 0.010	0.935 $\pm$ 0.010
	22	0.799 $\pm$ 0.014	0.882 $\pm$ 0.006	0.671 $\pm$ 0.019	0.855 $\pm$ 0.002	0.965 $\pm$ 0.006	0.918 $\pm$ 0.015
	23	0.757 $\pm$ 0.012	0.887 $\pm$ 0.006	0.577 $\pm$ 0.014	0.857 $\pm$ 0.007	0.989 $\pm$ 0.004	0.927 $\pm$ 0.011
	24	0.810 $\pm$ 0.014	0.893 $\pm$ 0.005	0.695 $\pm$ 0.021	0.861 $\pm$ 0.004	0.959 $\pm$ 0.003	0.935 $\pm$ 0.011
	25	0.732 $\pm$ 0.011	0.890 $\pm$ 0.009	0.527 $\pm$ 0.016	0.866 $\pm$ 0.006	0.997 $\pm$ 0.003	0.920 $\pm$ 0.014
	26	0.749 $\pm$ 0.013	0.895 $\pm$ 0.008	0.562 $\pm$ 0.017	0.858 $\pm$ 0.004	0.991 $\pm$ 0.003	0.943 $\pm$ 0.015
	27	0.812 $\pm$ 0.004	0.881 $\pm$ 0.005	0.741 $\pm$ 0.008	0.824 $\pm$ 0.006	0.903 $\pm$ 0.011	0.954 $\pm$ 0.011
	28	0.824 $\pm$ 0.010	0.871 $\pm$ 0.006	0.736 $\pm$ 0.015	0.802 $\pm$ 0.010	0.939 $\pm$ 0.005	0.961 $\pm$ 0.007
KaLM-mini	1	0.596 $\pm$ 0.012	0.893 $\pm$ 0.005	0.344 $\pm$ 0.003	0.846 $\pm$ 0.005	0.922 $\pm$ 0.022	0.953 $\pm$ 0.006

(continued on next page)

Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT)							
Model	Layer	Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
	2	0.620 $\pm$ 0.018	0.893 $\pm$ 0.003	0.422 $\pm$ 0.014	0.845 $\pm$ 0.005	0.876 $\pm$ 0.023	0.955 $\pm$ 0.008
	3	0.605 $\pm$ 0.011	0.893 $\pm$ 0.004	0.379 $\pm$ 0.008	0.843 $\pm$ 0.000	0.896 $\pm$ 0.018	0.959 $\pm$ 0.010
	4	0.562 $\pm$ 0.015	0.889 $\pm$ 0.005	0.265 $\pm$ 0.009	0.842 $\pm$ 0.008	0.945 $\pm$ 0.016	0.949 $\pm$ 0.008
	5	0.550 $\pm$ 0.013	0.890 $\pm$ 0.004	0.270 $\pm$ 0.005	0.841 $\pm$ 0.004	0.910 $\pm$ 0.019	0.952 $\pm$ 0.009
	6	0.522 $\pm$ 0.016	0.893 $\pm$ 0.004	0.203 $\pm$ 0.012	0.850 $\pm$ 0.005	0.934 $\pm$ 0.014	0.949 $\pm$ 0.014
	7	0.604 $\pm$ 0.018	0.893 $\pm$ 0.003	0.393 $\pm$ 0.018	0.840 $\pm$ 0.008	0.876 $\pm$ 0.023	0.962 $\pm$ 0.008
	8	0.607 $\pm$ 0.014	0.866 $\pm$ 0.003	0.388 $\pm$ 0.011	0.798 $\pm$ 0.008	0.891 $\pm$ 0.017	0.954 $\pm$ 0.007
	9	0.614 $\pm$ 0.011	0.874 $\pm$ 0.005	0.387 $\pm$ 0.008	0.821 $\pm$ 0.008	0.908 $\pm$ 0.013	0.943 $\pm$ 0.008
	10	0.602 $\pm$ 0.013	0.859 $\pm$ 0.008	0.360 $\pm$ 0.005	0.774 $\pm$ 0.010	0.916 $\pm$ 0.017	0.969 $\pm$ 0.010
	11	0.601 $\pm$ 0.012	0.864 $\pm$ 0.008	0.361 $\pm$ 0.007	0.792 $\pm$ 0.010	0.911 $\pm$ 0.020	0.956 $\pm$ 0.008
	12	0.640 $\pm$ 0.010	0.868 $\pm$ 0.006	0.416 $\pm$ 0.006	0.801 $\pm$ 0.007	0.929 $\pm$ 0.019	0.954 $\pm$ 0.006
	13	0.645 $\pm$ 0.011	0.862 $\pm$ 0.005	0.436 $\pm$ 0.005	0.793 $\pm$ 0.004	0.915 $\pm$ 0.018	0.951 $\pm$ 0.012
	14	0.654 $\pm$ 0.010	0.859 $\pm$ 0.006	0.479 $\pm$ 0.004	0.795 $\pm$ 0.008	0.880 $\pm$ 0.017	0.941 $\pm$ 0.011
	15	0.661 $\pm$ 0.009	0.870 $\pm$ 0.006	0.460 $\pm$ 0.009	0.814 $\pm$ 0.009	0.921 $\pm$ 0.014	0.941 $\pm$ 0.006
	16	0.685 $\pm$ 0.008	0.870 $\pm$ 0.007	0.477 $\pm$ 0.009	0.824 $\pm$ 0.008	0.953 $\pm$ 0.015	0.930 $\pm$ 0.008
	17	0.757 $\pm$ 0.006	0.886 $\pm$ 0.010	0.612 $\pm$ 0.016	0.841 $\pm$ 0.005	0.944 $\pm$ 0.010	0.943 $\pm$ 0.016
	18	0.781 $\pm$ 0.008	0.880 $\pm$ 0.008	0.636 $\pm$ 0.013	0.840 $\pm$ 0.002	0.967 $\pm$ 0.007	0.931 $\pm$ 0.016
	19	0.805 $\pm$ 0.012	0.877 $\pm$ 0.008	0.673 $\pm$ 0.020	0.831 $\pm$ 0.007	0.976 $\pm$ 0.003	0.935 $\pm$ 0.011
	20	0.794 $\pm$ 0.010	0.878 $\pm$ 0.006	0.650 $\pm$ 0.018	0.831 $\pm$ 0.003	0.979 $\pm$ 0.004	0.937 $\pm$ 0.011
	21	0.805 $\pm$ 0.010	0.871 $\pm$ 0.007	0.673 $\pm$ 0.016	0.831 $\pm$ 0.004	0.975 $\pm$ 0.006	0.924 $\pm$ 0.012
	22	0.871 $\pm$ 0.006	0.873 $\pm$ 0.006	0.811 $\pm$ 0.005	0.813 $\pm$ 0.007	0.949 $\pm$ 0.006	0.951 $\pm$ 0.007
	23	0.873 $\pm$ 0.007	0.885 $\pm$ 0.005	0.811 $\pm$ 0.006	0.835 $\pm$ 0.007	0.953 $\pm$ 0.007	0.950 $\pm$ 0.006
	24	0.848 $\pm$ 0.009	0.856 $\pm$ 0.004	0.749 $\pm$ 0.017	0.807 $\pm$ 0.006	0.975 $\pm$ 0.007	0.918 $\pm$ 0.009

G Reference Systems: Lexical Baselines and the Fine-Tuning Ceiling

Tables 2 and 3 carry four reference rows below the rule. Table 17 gives the three lexical baselines with their settings: the BM25 parameters, the character n -gram range, the train-only fitting, and the min-max rescaling that precedes the threshold fit. Table 18 gives the supervised fine-tuning ceiling with the pre-trained rows it is compared against.

The fine-tuned encoder is LaTa trained with symmetric InfoNCE over positive pairs and in-batch negatives (no two pairs from one directory in a batch), temperature 0.05, AdamW at learning rate 2×10^{-5} with weight decay 0.01, linear warmup over the first 10 percent of steps then linear decay, gradient clipping at 1.0, 16 pairs per batch, up to 8 epochs with early stopping after 3 epochs without dev improvement, and seed 42; the dev slice holds out 28 of the 190 train directories with at least two files (71 files), so training uses 499 of the 565 positive train pairs, and the checkpoint is chosen by dev directory accuracy at rank 1 with dev AUROC as the tie-break. The selected checkpoint is epoch 7, the last epoch run before early stopping fired, so the row is a ceiling at this training budget rather than an asymptote, and the ABTT sweep on the fine-tuned embeddings selects $D = 10$, the top of the grid, at every layer, so the tuned and fixed ABTT variants coincide there.

Baseline	Task A		Task B	
	AUROC	Gap	Assign.	Dir@1
BM25 (word)	0.974	0.243	62.5	56.3
TF-IDF char 3–5	0.987	0.555	91.8	89.9
Levenshtein	0.952	0.362	87.8	86.2

Table 17: Lexical baselines on the same 50/50 split, the same held-out test set and the same evaluation code as the embedding models, so the numbers are directly comparable to the headline Task A and Task B tables. Task A is measured over the test $n \times n$ score matrix: AUROC treats same-directory test pairs as positives, and Gap is the mean same-directory minus the mean different-directory score. Task B is reported in percent: Assign. is the existing-versus-new decision against a threshold τ fitted on train, and Dir@1 additionally requires the correct directory. TF-IDF char 3–5 is the strongest of the three at 89.9 percent Dir@1, which puts plain surface-form overlap within reach of the best embedding configurations on this corpus of hand-copied witnesses. BM25 separates pairs well but routes poorly: a raw BM25 score grows with query length, so each file’s maximum score tracks document length as much as relatedness, and the train-fitted τ transfers badly. BM25 uses $k_1 = 1.5$, $b = 0.75$ with document frequencies, vocabulary and average document length taken from train files only; the character TF-IDF vectoriser is likewise fitted on train only, and normalised Levenshtein has nothing to fit. Every score matrix is min-max rescaled using the train block; the map is monotone, so AUROC is unchanged, while the threshold τ is fitted on a fixed grid afterwards, which matters for BM25 (unrescaled, τ pins at the grid ceiling) and is the identity for char TF-IDF, whose train scores already span $[0, 1]$. Under the 5-seed query-versus-reference Task B protocol, in which only the query/reference partition of the test half is redrawn, directory accuracy at rank 1 is BM25 (word) 68.3 ± 7.1 ; TF-IDF char 3–5 91.1 ± 0.5 ; Levenshtein 87.2 ± 0.3 .

System	Task A		Task B	
	AUROC	Cosine gap	Assign. acc.	Dir. acc.@1
LaTa (pre-trained)	0.938 ₁₂	0.237 ₁₂	73.8 ₁	72.1 ₁
LaTa (pre-trained) + ABTT	0.971 ₁₂	0.525 ₁₂	88.5 ₈	86.1 ₈
LaTa (fine-tuned)	0.984 ₁₂	0.387 ₁₂	83.4 ₁₂	81.6 ₁₂
LaTa (fine-tuned) + ABTT	0.970 ₁₂	0.548 ₁₂	87.8 ₁₂	85.2 ₁₂

Table 18: Supervised fine-tuning reference ceiling on LaTa. The fine-tuned encoder is trained contrastively on the 565 positive pairs available in the train split (in-batch negatives, symmetric InfoNCE), with a dev slice carved out of train by directory for model selection; the test split is untouched until this evaluation. Task A columns are test AUROC and cosine gap at the layer chosen by train AUROC; Task B columns are test assignment accuracy and directory accuracy at rank 1, in percent, at the layer chosen by train directory accuracy, with τ learned on train. Layer indices are subscripts. ABTT rows sweep D per layer on train and select $D = 10$ everywhere, which is the top of the grid $\{1, 2, 3, 5, 7, 10\}$, so the tuned and fixed variants coincide. The selected checkpoint is epoch 7, the terminal epoch of the sweep under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. This is a reference ceiling, not a proposed method: it consumes supervision the zero-shot pipeline never sees.

H Secondary Attribution Metrics and Sweeps

Ranking conventions. Two ranking conventions hold for every curve metric and every method, variant and control alike. Tokens are ranked by attribution magnitude $|a|$ rather than by signed importance, because the question is which tokens carry the pair cosine and both methods produce two-signed scores. Ties are broken by token position, which matters most for MaRC, whose many exactly-zero scores are then ordered by reading order. Absolute values are therefore not comparable with a signed-ranking implementation, and Kendall τ_b in Table 19 is the tie-corrected companion to ρ_{LOO} under the second convention.

Secondary metrics. Figure 4 plots the rank-faithfulness column of Table 4. Table 19 carries the metrics the main table does not: tie-corrected Kendall τ_b , chance-corrected insertion faithfulness, and the three ERASER-style rationale metrics at the global threshold settings. Five computed quantities are printed in neither table: the raw deletion and insertion areas, which appear here only as the chance-corrected gaps; AOPC-Sufficiency and AOPC-Comprehensiveness, whose per-threshold components are in the sweeps below; and the per-cell shuffled-attribution gaps, which the main text reports as a validity statement rather than as a result column. The rest of this appendix expands those rationale metrics across Sufficiency and Comprehensiveness thresholds of 10%, 25%, and 50%, and MinFrac recovery thresholds of 0.70, 0.80, 0.90, and 0.95. We use one global midpoint setting rather than selecting thresholds per model or per method. Table 20 gives the sweep for integrated gradients and retrieval-adapted MaRC, and Table 21 gives it for the supplemental attribution methods. Every table in this appendix uses the same representation-level erasure operator as the main table, so the cells are comparable across them.

I Qualitative Attribution Examples

This appendix keeps the example-level views for one PhilTa retrieval pair. Figure 5 gives the integrated-gradient pair matrix before and after ABTT. The attention map in Figure 6 is context only, since ABTT is applied after encoding. The retrieval-adapted MaRC masks in Figure 7 show the learned-mask view for the same pair.

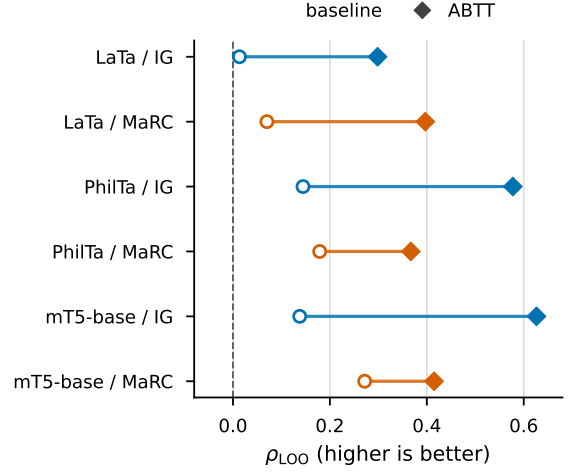


Figure 4: ρ_{LOO} foreground view for the main candidate-attribution result. Each line connects the baseline and ABTT variants for one model-method cell at the predeclared operational attribution layer. ABTT improves the leave-one-out rank-correlation signal in all six cells; Table 19 shows the secondary metrics which qualify this narrower faithfulness claim.

J Cluster Geometry

Figures 8 and 9 give the t-SNE and UMAP panels for LaTa, PhilTa, and mT5-base, and Figures 10 and 11 extend them to LaBSE, Qwen3-0.6B, and KaLM-mini. All four use the same six directories and the same baseline versus ABTT-only comparison.

Model	Method	τ_{LOO}		InsAUC gap		Suff@25%		Comp@25%		MinFrac@0.80	
		base	ABTT	base	ABTT	base	ABTT	base	ABTT	base	ABTT
LaTa	IG	0.007	0.212	0.153	0.032	0.965	0.729	0.658	0.194	0.042	0.330
	MaRC	0.059	0.292	-0.201	0.098	0.556	0.847	0.329	0.272	0.398	0.291
PhilTa	IG	0.106	0.432	0.118	0.244	0.965	0.941	0.988	0.546	0.031	0.144
	MaRC	0.133	0.258	0.104	0.200	0.952	0.883	0.951	0.443	0.058	0.296
mT5-base	IG	0.102	0.465	-0.028	0.353	0.885	1.007	0.055	0.653	0.134	0.232
	MaRC	0.186	0.291	0.017	0.243	0.936	0.840	0.047	0.591	0.032	0.340

Table 19: Secondary attribution metrics, on the same pairs, the same layers and the same erasure operator as Table 4. Boldface marks the better variant within a pair; higher is better everywhere except MinFrac. None of these columns is in the main table, and each is out for its own reason. Kendall τ_b agrees with ρ_{LOO} in 6/6 cells, but it is the tie-corrected twin of the same statistic rather than a second witness. Chance-corrected insertion faithfulness favours ABTT in 5/6 cells, and we do not report it in the main table because in two of the twelve cells, both of them baseline cells, the real attribution does not beat a permutation of its own scores, so the measurement does not meet the validity bar we set for a headline column. The threshold-based ERASER metrics are reported for completeness: their baseline-versus-ABTT verdict depends on the threshold and on the erasure operator, which is why the main table uses threshold-free, chance-corrected metrics instead. The InsAUC columns average 191 to 195 ABTT pairs against the baseline’s 200, for the same small-denominator reason. Full threshold sweeps are in Tables 20 and 21.

Model	Method	$\rho_{LOO} \uparrow$		Suff@10% $\uparrow$		Suff@25% $\uparrow$		Suff@50% $\uparrow$		Comp@10% $\uparrow$		Comp@25% $\uparrow$		Comp@50% $\uparrow$		MinFrac@0.70 $\downarrow$		MinFrac@0.80 $\downarrow$		MinFrac@0.90 $\downarrow$		MinFrac@0.95 $\downarrow$	
		base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt
LaTa	IG	0.013	0.298	0.949	0.483	0.965	0.729	0.970	0.923	0.304	0.076	0.658	0.194	0.802	0.372	0.040	0.243	0.042	0.330	0.045	0.456	0.057	0.539
	MaRC	0.070	0.397	0.419	0.723	0.556	0.847	0.603	0.931	0.114	0.107	0.329	0.272	0.466	0.519	0.375	0.145	0.398	0.291	0.423	0.447	0.455	0.539
PhilTa	IG	0.144	0.577	0.946	0.832	0.965	0.941	0.983	1.030	0.702	0.301	0.988	0.546	1.026	0.783	0.028	0.088	0.031	0.144	0.045	0.242	0.084	0.319
	MaRC	0.179	0.367	0.913	0.671	0.952	0.883	0.953	0.997	0.649	0.199	0.951	0.443	0.971	0.692	0.058	0.216	0.058	0.296	0.060	0.386	0.075	0.451
mT5-base	IG	0.138	0.626	0.788	0.807	0.885	1.007	0.937	1.148	0.013	0.328	0.055	0.653	0.275	0.999	0.067	0.172	0.134	0.232	0.342	0.317	0.605	0.368
	MaRC	0.272	0.415	0.907	0.660	0.936	0.840	0.963	1.017	0.011	0.284	0.047	0.591	0.139	0.880	0.015	0.261	0.032	0.340	0.094	0.441	0.231	0.500

Table 20: Appendix attribution sweep for the two paper-facing attribution views, IG and retrieval-adapted MARC, across the three main-paper models. Direction arrows are printed in the column headers. Sufficiency and Comprehensiveness are ratio metrics at {10%, 25%, 50%}; MinFrac is the minimum retained query-token fraction needed to recover $\tau \in \{0.70, 0.80, 0.90, 0.95\}$ of the full cosine (lower is better). Ratio metrics require $|S_v(q_{full}, c)| \geq 0.05$; ρ_{LOO} excludes $|\Delta| < 10^{-6}$. The completeness report contains no missing metric cells.

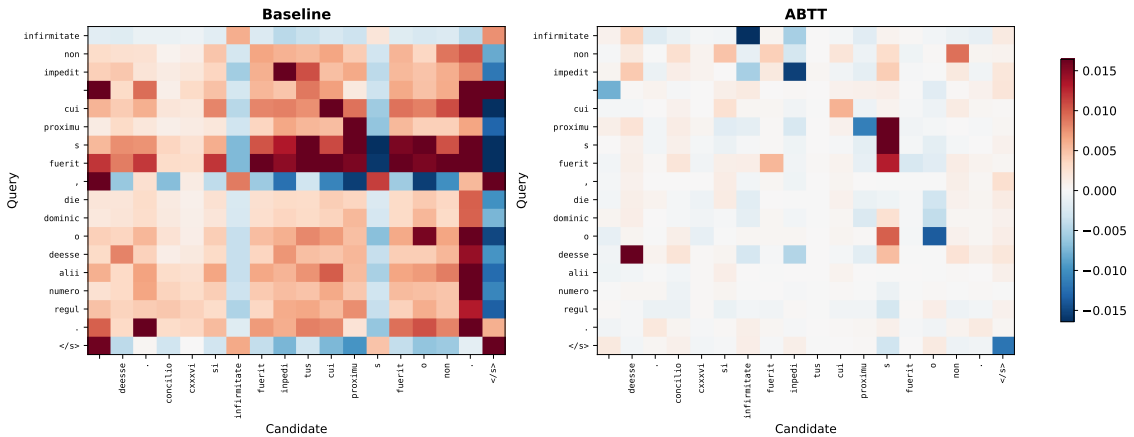


Figure 5: Derived pairwise heatmap for one PhilTa retrieval case. The left panel shows the baseline representation and the right panel shows the same example after ABTT. Both axes use original decoded tokens. Cell values combine token-token cosine similarity with single-sequence integrated-gradient scores.

Model	Method	$\rho_{\text{LOO}} \uparrow$		Suff@10% $\uparrow$		Suff@25% $\uparrow$		Suff@50% $\uparrow$		Comp@10% $\uparrow$		Comp@25% $\uparrow$		Comp@50% $\uparrow$		MinFrac@0.70 $\downarrow$		MinFrac@0.80 $\downarrow$		MinFrac@0.90 $\downarrow$		MinFrac@0.95 $\downarrow$	
		base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt
LaTa	BERTScore	0.227	0.397	1.040	0.483	1.048	0.707	1.048	0.890	0.909	0.089	1.205	0.216	1.189	0.426	0.018	0.257	0.020	0.344	0.024	0.491	0.033	0.583
	OT	0.013	0.298	0.949	0.483	0.965	0.729	0.970	0.923	0.304	0.076	0.658	0.194	0.802	0.372	0.040	0.243	0.042	0.330	0.045	0.456	0.057	0.539
	Att-weighted	-0.175	0.069	-0.562	0.277	-0.290	0.466	-0.056	0.707	0.003	0.060	-0.003	0.163	-0.018	0.296	0.749	0.468	0.759	0.572	0.774	0.696	0.795	0.772
	Att-standalone	0.046	0.483	0.152	0.568	0.268	0.775	0.429	0.944	0.034	0.144	0.091	0.315	0.305	0.563	0.546	0.216	0.560	0.314	0.584	0.436	0.600	0.526
	DLA	0.029	0.427	1.013	0.548	0.998	0.758	0.999	0.926	0.867	0.150	1.055	0.303	1.051	0.510	0.017	0.223	0.020	0.321	0.026	0.463	0.036	0.548
	random	-0.002	-0.008	0.463	0.524	0.731	0.701	0.905	0.854	0.025	0.025	0.049	0.058	0.108	0.145	0.179	0.245	0.184	0.358	0.197	0.539	0.214	0.667
	inverse	-0.013	-0.298	0.077	0.283	0.199	0.444	0.198	0.635	0.000	-0.007	0.003	-0.000	0.030	0.081	0.587	0.541	0.599	0.674	0.626	0.811	0.652	0.884
PhiTa	BERTScore	0.357	0.581	1.004	0.824	1.029	0.962	1.047	1.030	0.681	0.251	1.022	0.480	1.038	0.700	0.019	0.114	0.021	0.177	0.034	0.277	0.069	0.338
	OT	0.144	0.577	0.946	0.832	0.965	0.941	0.983	1.030	0.702	0.301	0.988	0.546	1.026	0.783	0.028	0.088	0.031	0.144	0.045	0.242	0.084	0.319
	Att-weighted	0.068	0.088	0.108	0.398	0.176	0.628	0.226	0.782	0.003	0.066	0.013	0.198	0.033	0.377	0.659	0.361	0.671	0.473	0.690	0.624	0.716	0.707
	Att-standalone	0.114	0.196	-0.020	0.453	0.020	0.696	0.045	0.842	0.006	0.105	0.037	0.272	0.059	0.490	0.768	0.288	0.775	0.381	0.791	0.513	0.801	0.599
	DLA	0.231	0.541	0.946	0.883	0.965	0.973	0.988	1.002	0.709	0.345	1.009	0.608	1.040	0.808	0.029	0.070	0.031	0.119	0.042	0.215	0.072	0.300
	random	0.001	-0.008	0.623	0.381	0.855	0.574	0.964	0.779	0.010	0.033	0.013	0.091	0.053	0.226	0.131	0.359	0.133	0.489	0.145	0.657	0.168	0.763
	inverse	-0.144	-0.577	-0.053	0.020	-0.044	0.086	-0.025	0.223	0.001	-0.035	0.005	-0.045	0.017	-0.028	0.910	0.868	0.918	0.913	0.930	0.954	0.940	0.972
mT5-base	BERTScore	0.280	0.550	0.311	0.715	0.591	0.946	0.863	1.108	0.012	0.234	0.031	0.514	0.069	0.832	0.211	0.198	0.270	0.261	0.385	0.334	0.489	0.377
	OT	0.138	0.626	0.788	0.807	0.885	1.007	0.937	1.148	0.013	0.328	0.055	0.653	0.275	0.999	0.067	0.172	0.134	0.232	0.342	0.317	0.605	0.368
	Att-weighted	0.184	0.564	0.913	0.908	0.941	0.999	0.947	1.081	0.011	0.333	0.049	0.634	0.284	0.926	0.021	0.103	0.043	0.171	0.167	0.270	0.441	0.338
	Att-standalone	0.099	0.044	0.508	0.181	0.486	0.386	0.818	0.590	0.002	0.197	0.024	0.335	0.087	0.403	0.050	0.526	0.137	0.635	0.423	0.741	0.606	0.815
	DLA	0.132	0.491	-0.282	0.927	0.321	0.966	0.875	0.971	0.027	0.370	0.036	0.630	0.055	0.844	0.359	0.093	0.402	0.163	0.481	0.283	0.564	0.366
	random	-0.000	-0.005	0.832	0.340	0.927	0.528	0.971	0.725	0.004	0.052	0.011	0.132	0.030	0.283	0.040	0.422	0.068	0.560	0.146	0.716	0.263	0.803
	inverse	-0.138	-0.626	-0.266	-0.219	0.098	-0.197	0.734	0.010	0.018	-0.099	0.041	-0.167	0.064	-0.144	0.446	0.872	0.520	0.916	0.622	0.959	0.712	0.979

Table 21: Supplemental attribution sweep for additional methods and sanity baselines. *random* and *inverse* are diagnostic baselines, not paper-facing attribution methods. Direction arrows are printed in the column headers. Sufficiency and Comprehensiveness are ratio metrics at $\{10\%, 25\%, 50\%\}$; MinFrac is the minimum retained query-token fraction needed to recover $\tau \in \{0.70, 0.80, 0.90, 0.95\}$ of the full cosine (lower is better). Ratio metrics require $|S_v(q_{\text{full}}, c)| \geq 0.05$; ρ_{LOO} excludes $|\Delta| < 10^{-6}$. The completeness report contains no missing metric cells.

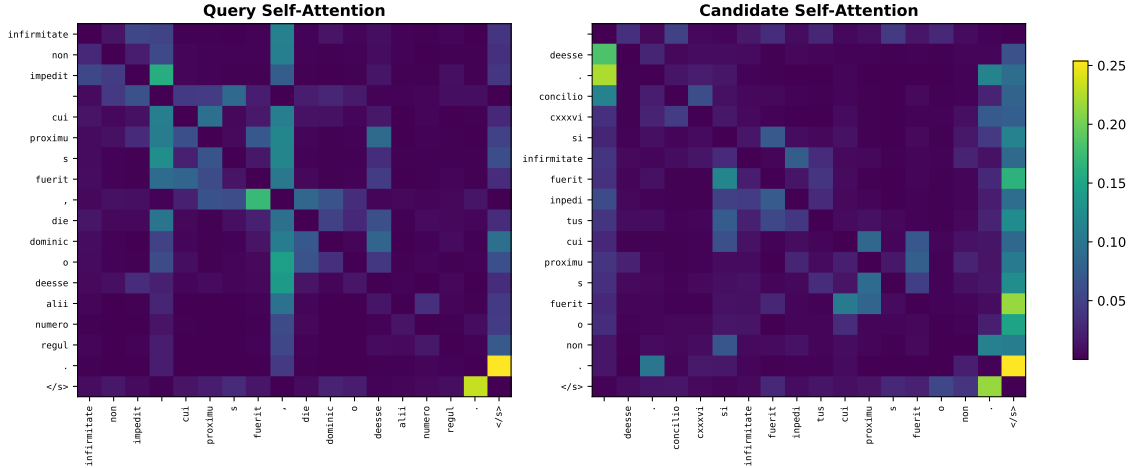


Figure 6: Companion self-attention maps for the same PhiTa example as Figure 5. ABTT is applied after encoding, so attention is included only as supporting context.

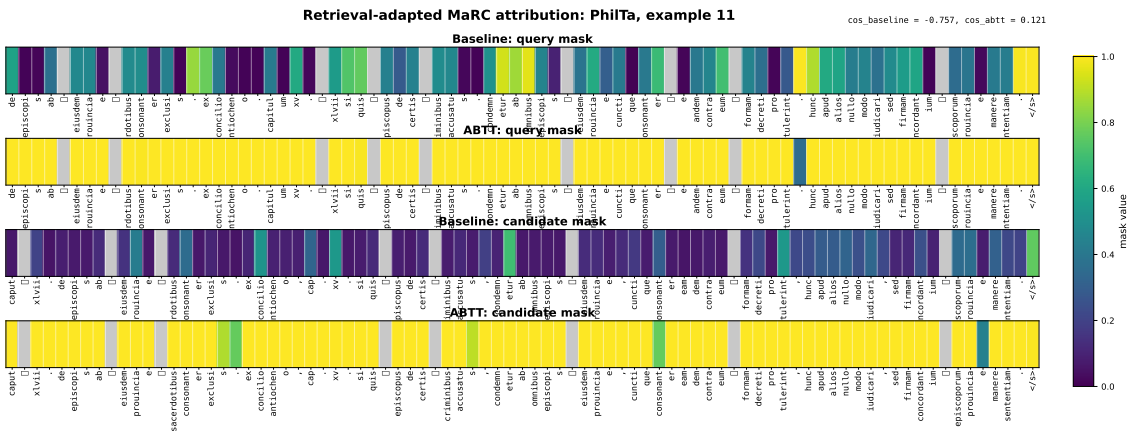


Figure 7: Retrieval-adapted MaRC masks for the same PhiTa pair as Figure 5. Top row: mask on the query with the candidate fixed. Bottom row: mask on the candidate with the query fixed. Left column: baseline cosine target. Right column: ABTT-corrected cosine target.

2D TSNE of labelled fragments at each model's best layer

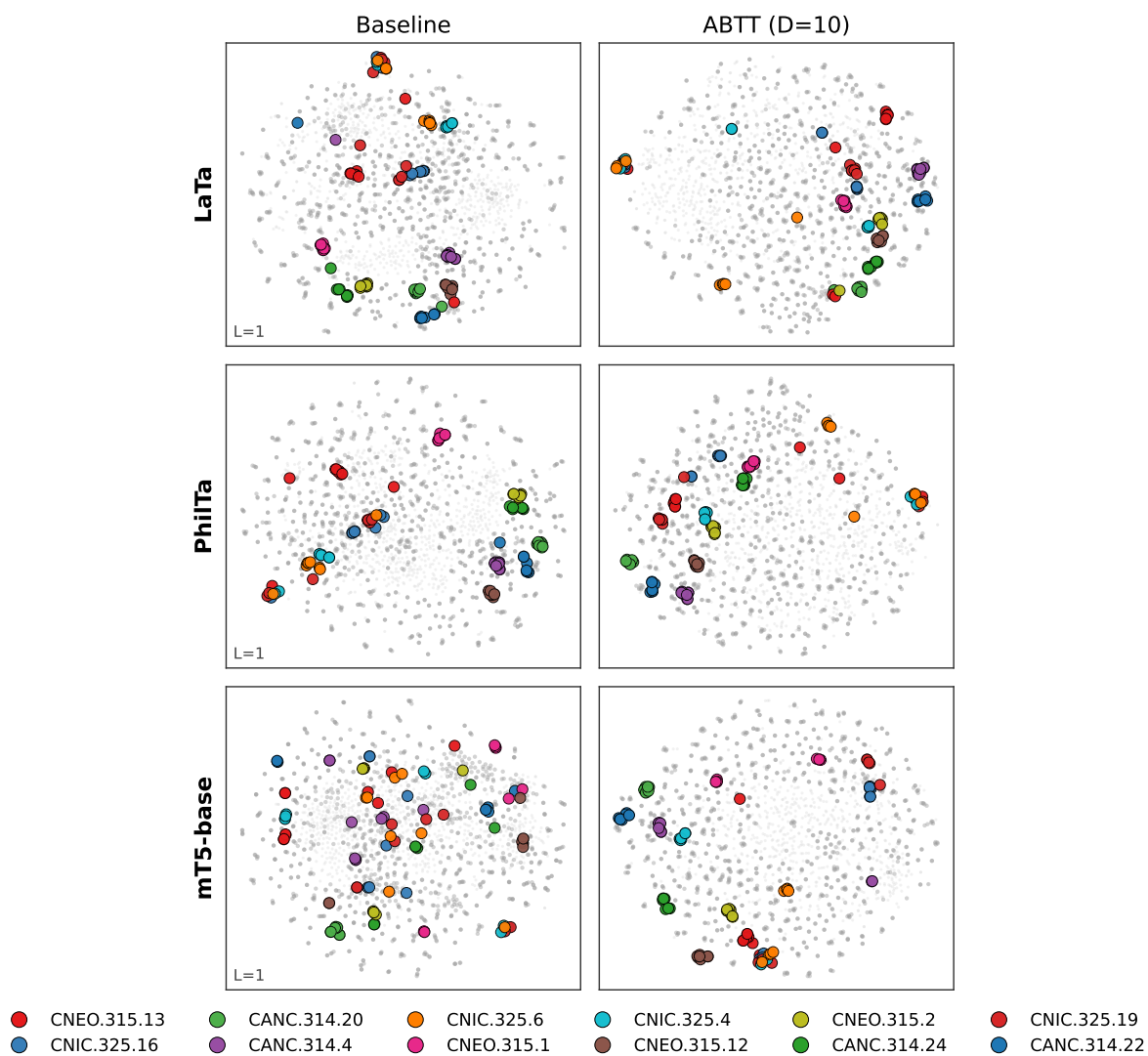


Figure 8: t-SNE projections of the 1,705 manuscript witnesses at each model's selected layer for LaTa, PhilTa, and mT5-base. Columns compare baseline mean pooling with ABTT-only using $D = 10$. Colors mark the six most-populated directories.

2D UMAP of labelled fragments at each model's best layer

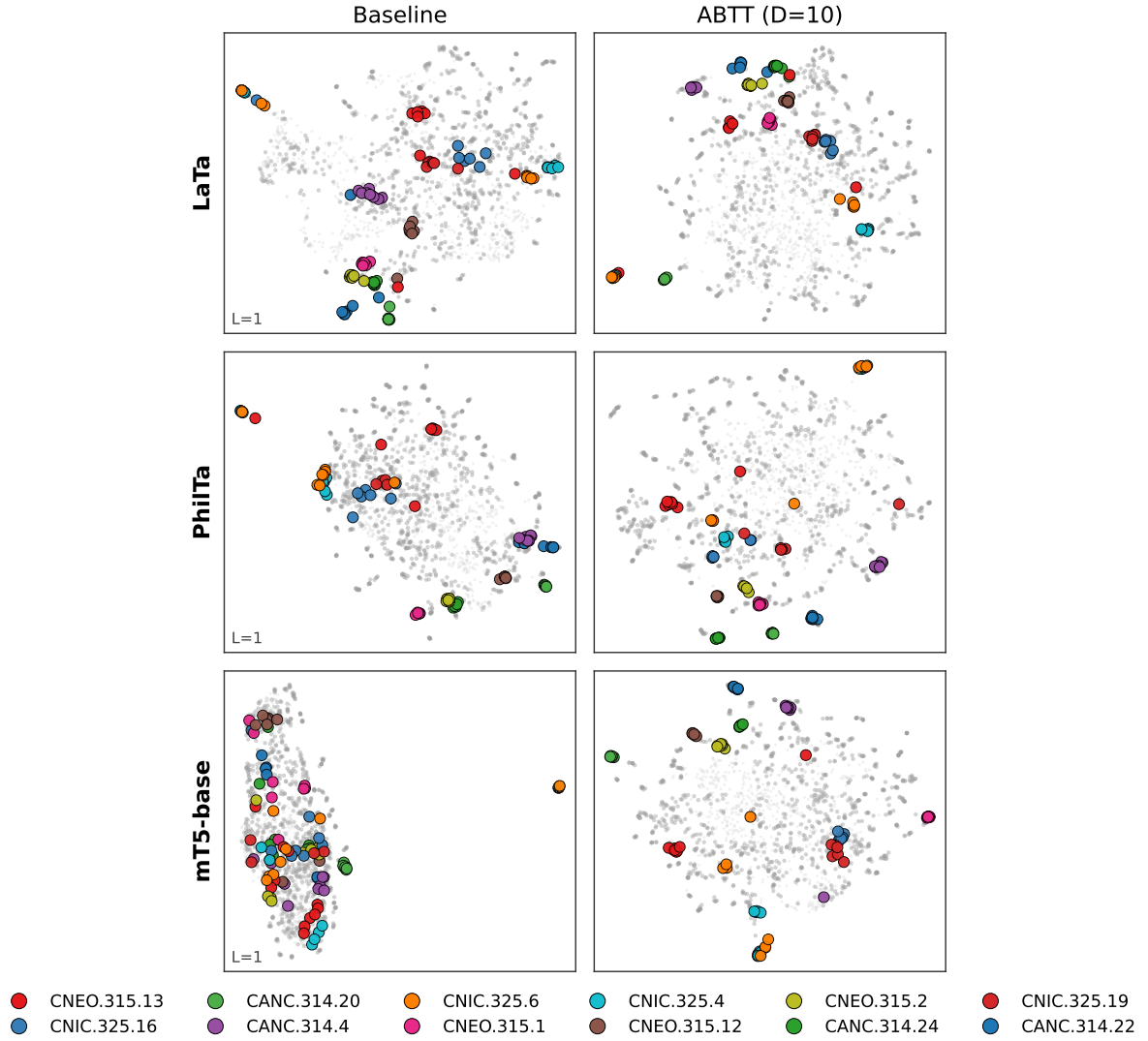


Figure 9: UMAP projections with the same configuration as Figure 8. The same cluster-tightening pattern appears under a different projection method.

2D TSNE of labelled fragments at each model's best layer

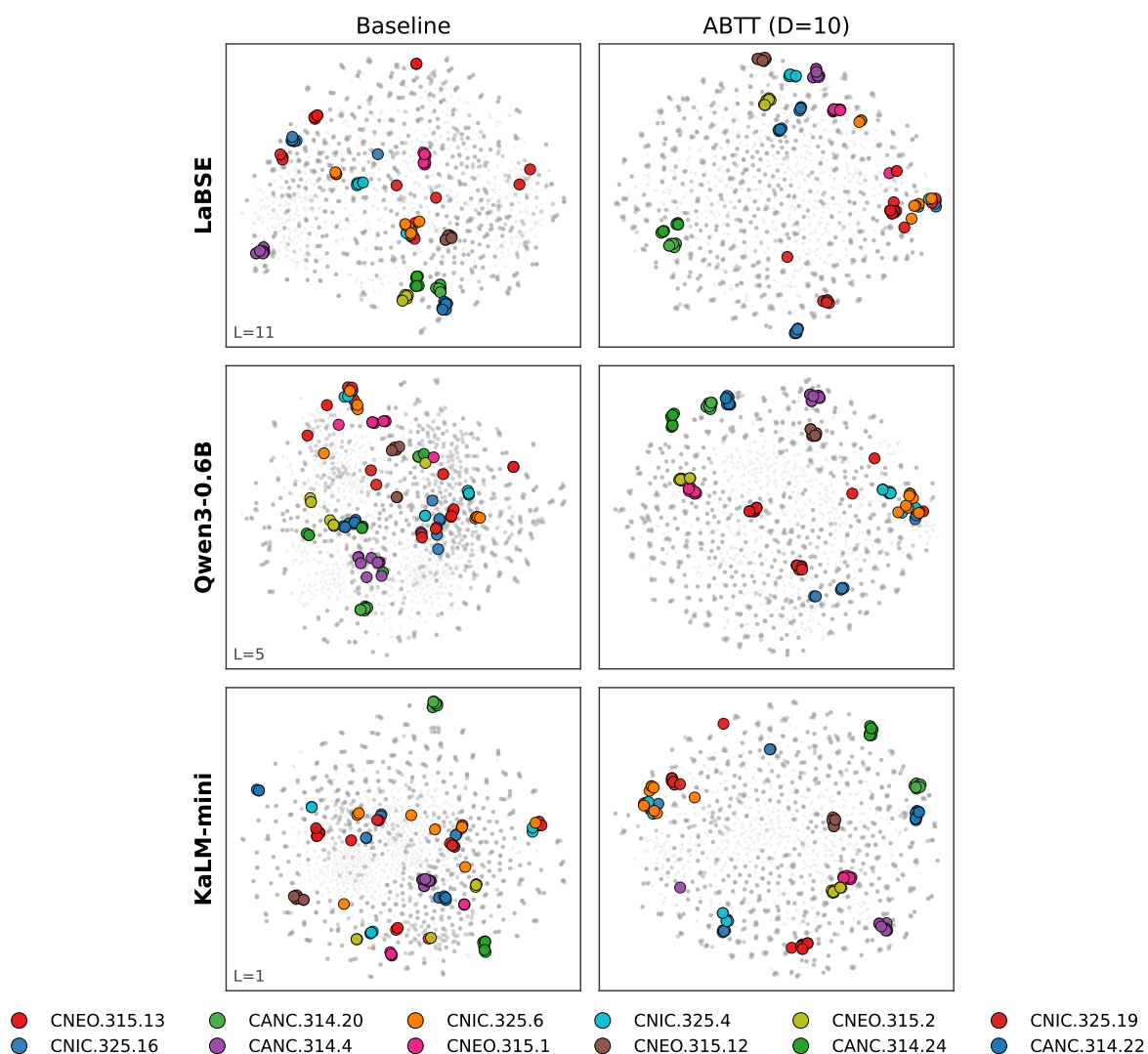


Figure 10: t-SNE projections of the six most-populated directories at each selected layer for LaBSE, Qwen3-0.6B, and KaLM-mini. Columns compare baseline mean pooling with ABTT-only using $D = 10$.

2D UMAP of labelled fragments at each model's best layer

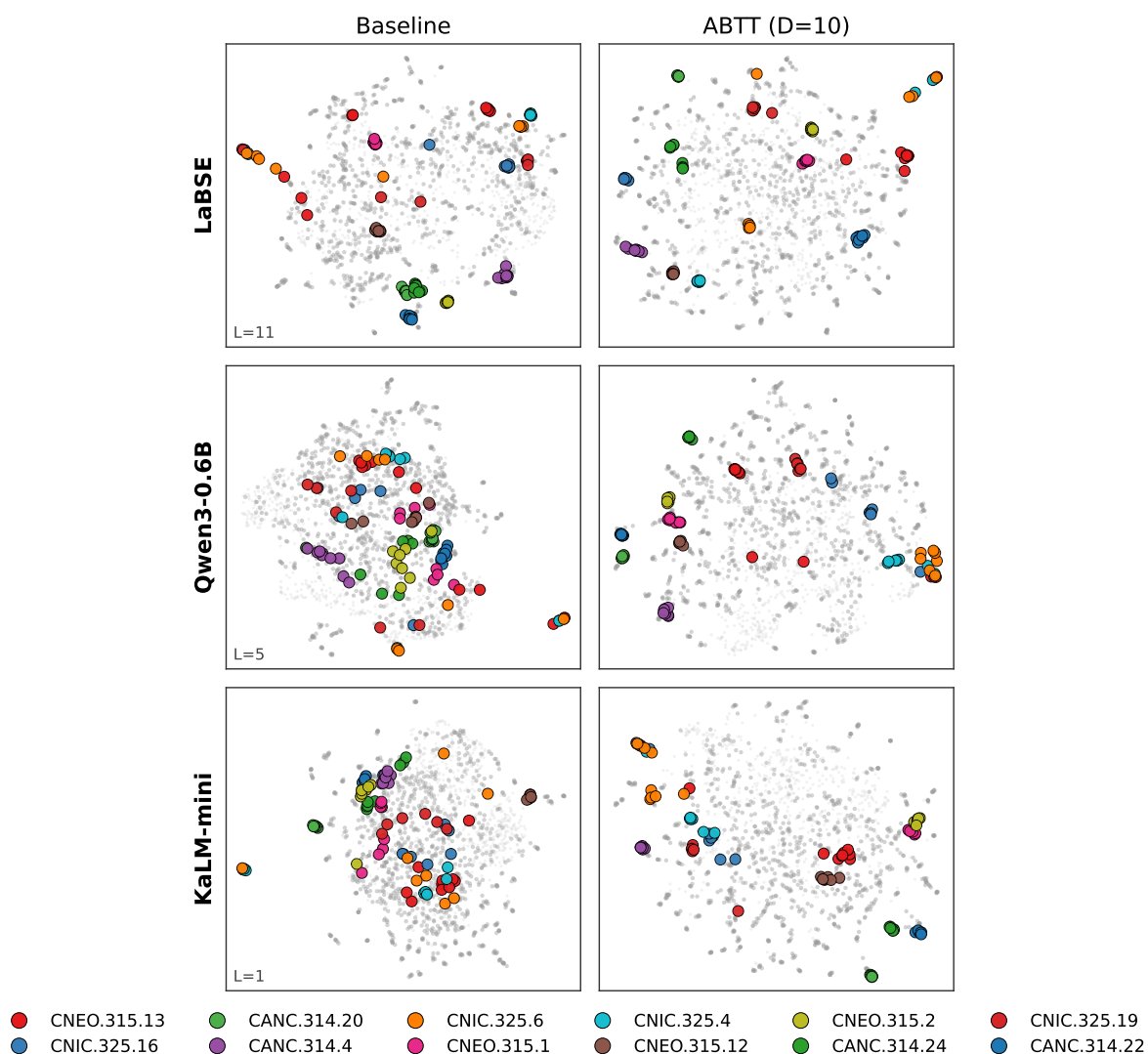


Figure 11: UMAP projections matching Figure 10. The cluster-tightening pattern appears under a second projection method.